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Collaborative Untethered Transport of Free-Floating Objects in Microgravity

A Rendezvous-Based Planning and Control Framework

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Abstract

Collaborative transport of passive objects in microgravity is a key capability for emerging space applications such as on-orbit servicing, debris removal, and in-space assembly. Traditional approaches largely rely on rigid connections, which typically require dedicated infrastructure to be present on the object, or on manipulators, in which the robot and object are mechanically coupled before transport commences. Such methods are typically costly and limited in scalability and reusability, as they require precise and often complex docking procedures prior to transport. More recently, impact-based approaches have been proposed to enable untethered collaborative transport, eliminating the need for rigid attachments.

This thesis presents a formal planning and control framework for untethered collaborative transport of a free-floating, non-cooperative object in microgravity. The proposed method replaces impulsive interactions with rendezvous-preceded, untethered contacts, enabling gentle momentum transfer while maintaining stability and safety. High-level task requirements and safety constraints are formally specified using Signal Temporal Logic and encoded into a Mixed-Integer Convex Programming problem.

The resulting framework follows a hierarchical structure consisting of a Mixed-Integer Programming–based offline planner, an online re-planner that enforces geometric feasibility, and a Model Predictive Control layer for trajectory tracking and interaction execution. The approach is evaluated in a two-dimensional microgravity analogue environment using high-fidelity simulations of free-flying robots and objects.

The results indicate that rendezvous-based interactions enable collaborative, untethered transport with low interaction forces and adequate accuracy over short- to medium-duration planning horizons. Beyond demonstrating feasibility, the proposed framework provides a systematic way to formally specify, plan, and execute untethered interactions in two-dimensional microgravity analogue environments, supporting future autonomous space operations. Code and videos are available at `https://arian-kourangi.github.io/DA236X-Master-Thesis/`

Keywords

Collaborative, Untethered, Transport, Microgravity, Mixed-Integer Programming, Rendezvous, Signal Temporal Logic

Sammanfattning

Kollaborativ transport av passiva objekt i mikrogravitation är en central förmåga för framväxande rymdtillämpningar såsom service i omloppsbana, rymdskrotsborttagning och montering av strukturer i rymden. Traditionella metoder förlitar sig i stor utsträckning på fasta kopplingar, vilket ofta kräver att särskild infrastruktur finns tillgänglig på objektet, eller på manipulatorer där robot och objekt mekaniskt kopplas samman innan transporten påbörjas. Dessa metoder är i regel kostsamma och begränsade när det gäller skalbarhet och återanvändbarhet, eftersom de förutsätter hög precision och ofta komplexa dockningsförfaranden innan transport. På senare tid har därför kollisionsbaserade metoder föreslagits för att möjliggöra obunden kollaborativ transport, vilket eliminerar behovet av fasta kopplingar.

Denna uppsats presenterar ett formellt planerings- och regleringsramverk för obunden kollaborativ transport av ett fritt svävande, icke samverkande objekt i mikrogravitation. Den föreslagna metoden ersätter impulsiva interaktioner med rendezvous-föregångna, obundna kontakter, vilket möjliggör skonsam överföring av momentum samtidigt som stabilitet och säkerhet bibehålls. Övergripande uppgiftskrav och säkerhetsrestriktioner specificeras formellt med hjälp av Signal Temporal Logic och kodas i ett Mixed-Integer Convex Programming-problem.

Det resulterande ramverket har en hierarkisk struktur som består av en Mixed-Integer Programming-baserad offlineplanerare, en onlineomplanerare som säkerställer geometrisk genomförbarhet samt en Model Predictive Control-regulator för trajektoriespårning och genomförande av interaktioner. Metoden utvärderas i en tvådimensionell mikrogravitationsanalog miljö med hjälp av detaljerade simuleringar av fritt flygande robotar och objekt.

Resultaten visar att rendezvous-baserade interaktioner möjliggör kollaborativ, obunden transport med låga interaktionskrafter och tillräcklig noggrannhet över korta till medellånga planeringshorisonter. Utöver att påvisa genomförbarhet tillhandahåller det föreslagna ramverket ett systematiskt sätt att formellt specificera, planera och genomföra obundna interaktioner i tvådimensionella mikrogravitationsanaloga miljöer, vilket kan stödja framtida autonoma rymdoperationer. Kod och videor finns tillgängliga på `https://arian-kourangi.github.io/DA236X-Master-Thesis/`

Nyckelord

Kollaborativt, Obunden, Transport, Mikrogravitation, Mixed-Integer programmering, Rendezvous, Signal Temporal Logik

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Stockholm, April 2026
Arian Kourangi Esfahani

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List of acronyms and abbreviations

ATMOS	Autonomy Testbed for Multi-purpose Orbiting Systems
CBF	Control Barrier Function
KTH	KTH Royal Institute of Technology
MICP	Mixed Integer Convex Programming
MILP	Mixed Integer Linear Programming
MIP	Mixed Integer Programming
MPC	Model Predictive Control
QP	Quadratic Program
RoI	Region of Interest
STL	Signal Temporal Logic

Chapter 1

Introduction

1.1 Background

Transporting passive objects or non-cooperative agents in the microgravity of space is the subject of extensive scientific research. Such transport capabilities are fundamental for emerging space activities, including satellite servicing, debris removal, and the assembly of large infrastructure like solar power stations and telescopes [1, 2]. Unlike terrestrial transport, there are no natural stabilizing forces such as gravity or friction. Once an object is set in motion, it will not naturally come to rest, making even simple manoeuvres highly non-trivial.

A large body of work exists on transporting non-cooperative objects in microgravity [3–5], ranging from single-robot [6] to multi-agent [7] solutions, and using various control schemes and collaboration methods to transport objects rigidly anchored to the robots [8, 9]. Furthermore, to facilitate transport when robots are not physically connected to objects, several studies focus on trajectory planning and impact kinematics for capturing objects with manipulators[10, 11]. Flores-Abad *et al*. [3] and Fallahiarezoodar *et al*. [4] review multiple strategies for on-orbit service, highlighting the need for careful planning and control during robot-object interactions to ensure safe and reliable docking or capturing.

Collaborative transport is particularly attractive because it allows multiple small robots to replace a single heavy manipulator system [12], which is typically more costly to deploy and performs slower than collaborative systems. However, coordinating multiple free-flying agents introduces challenges such as coupled dynamics, synchronization of contact forces, and stability of the object's trajectory [13].

In the realm of in-space manipulation and transport, an important distinction is between rigidly attached transport, and untethered or contact-based interactions. The advantage of rigid attachments lies in the ability to model the combined system as a single rigid body, significantly simplifying control and enabling precise trajectory tracking and force regulation using conventional model-based controllers [3, 5]. This approach however, relies on accurate docking or grasping procedures, which typically require in-depth knowledge of the targets inertial properties and fine-grained control. In contrast, untethered manipulation avoids permanent physical coupling and reduces structural and docking complexity, but leads to highly coupled, under-actuated dynamics where momentum exchange must be carefully managed, making stability and safety significantly more challenging [4].

1.1.1 Limitations of Existing Approaches

As noted in the aforementioned studies, most transportation methods in microgravity rely on rigid connections, tethers, or manipulators. To the best of our knowledge, research in purely untethered transport remains a limited body of work, with most active research focusing on strategies that rely on either docking or capture before actively transporting or manipulating the object. In a more recent study, Verhagen and Tumova [14] demonstrate a method for collaborative untethered transportation via impact interactions in microgravity, utilizing Signal Temporal Logic (STL) and Mixed Integer Programming (MIP) to formalize constraints and interactions between robots and objects to achieve the desired transport. In this context, untethered entails the absence of any rigid attachments, grasping manipulators, or tethers between agents and objects. This method eliminates the need for rigid anchoring or capture procedures using manipulators, reducing system complexity and deployment costs.

Using impacts in this way introduces other concerns, particularly the risk of damage to the robot or object during a collision. It also increases the danger of contributing to the growing problem of space debris. Another issue is the inherent under-actuation of the object. Without a rigid connection, slipping can occur, making it very difficult to control orientation or rotational velocity. This lack of full control authority not only reduces precision but also increases the likelihood of unstable or unsafe trajectories.

1.1.2 Motivation for Untethered Transport

Untethered transport offers several practical advantages over rigidly attached manipulation in space systems. By avoiding permanent physical coupling, servicing robots do not need to model or control the full dynamics of a combined robot–object system, which reduces controller complexity and sensitivity to uncertainty in the target's inertial properties. It also relaxes mechanical and structural requirements, since target objects do not need dedicated docking interfaces or purpose-built agents. This lowers system cost and supports scalable operation across diverse targets without redesign. As a result, untethered transport is particularly well suited for large-scale applications such as debris mitigation, in-orbit assembly, and distributed logistics.

For the long-term viability of untethered collaborative transport, a shift towards gentle, soft-contact interactions is required. In the space domain, this problem is closely related to the well-studied challenge of rendezvous, where two objects, such as satellites, must achieve simultaneous consensus in both position and velocity before docking or interaction can occur. Unlike terrestrial or aerial robotics, rendezvous in microgravity is particularly demanding. Objects may possess a large angular momentum, relative motion must be estimated under significant sensing uncertainty, and even minor mismatches can lead to destabilization or mission failure [3]. The study of automated control strategies for rendezvous has therefore been the focus of a large body of research. For example, [15] provides a comprehensive overview of classical and modern approaches, highlighting the persistent challenges of robustness, safety, and efficiency.

Even assuming a successful rendezvous, the subsequent interaction phase introduces additional challenges. Since the object is not physically constrained, forces can only be applied through a single contact point during interaction. Consequently, the direction of the applied force must remain approximately constant throughout contact, severely restricting the set of feasible manoeuvrers and making it difficult to control both the translational and rotational motion of the object simultaneously. Therefore, achieving precise and stable transport under these conditions requires coordination and planning that explicitly considers these physical limitations and constraints. To further clarify the distinctions between the interaction paradigms discussed above, Fig. 1.1 provides a conceptual comparison of rigidly docked transport, impact-based transport, and rendezvous-preceded interactions.

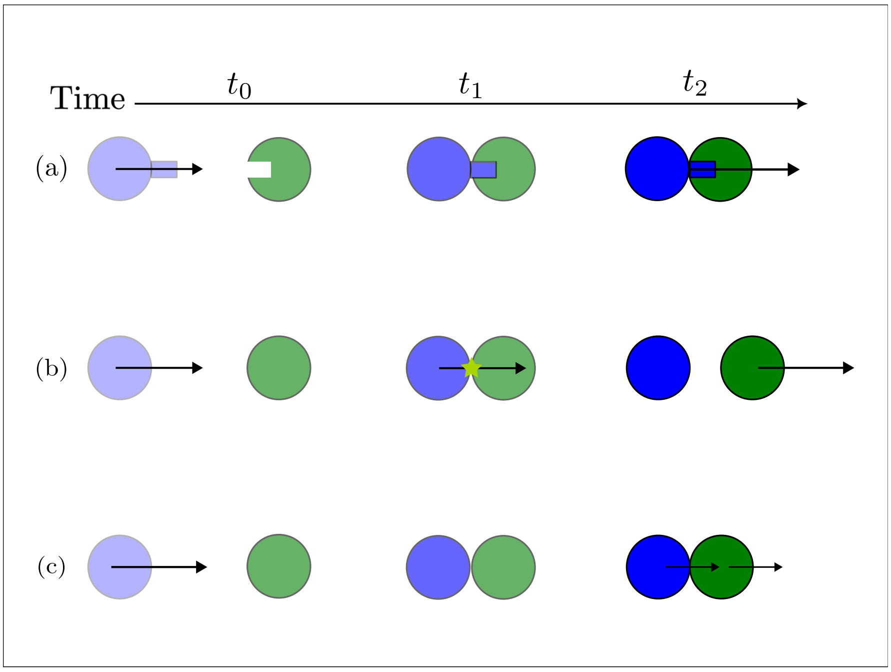

Figure 1.1: Conceptual comparison of interaction paradigms for transport in microgravity. (a) Rigid docking, where robots (shown in blue) and objects (shown in green) are mechanically coupled prior to transport. (b) Impact-based transport, in which momentum is transferred impulsively through brief contact. (c) Rendezvous-preceded compliant interaction, where untethered contact enables controlled momentum transfer while limiting interaction forces. Arrows indicate translational velocity.

1.1.3 Formal Methods for Multi-Agent Task Planning

The physical constraints and long interaction horizons of untethered collaborative transport make planning particularly challenging. In related settings, many existing works have turned to formal specification frameworks, such as STL, to reason about safety and interaction constraints in complex multi-agent systems. STL allows for the encoding of high-level temporal and safety constraints into mathematically precise conditions [16]. This provides a systematic way to design planners and task-level synthesis methods with formal guarantees on safety and correctness, which is crucial in space missions where failure has irrecoverable consequences. However, STL synthesis faces well-known challenges in scalability with respect to task complexity and planning horizon [17]. Consequently, much of the recent scientific effort has focused on extending STL to handle more complex task descriptions, multi-agent collaboration, and online execution. For example, Lindemann and Dimarogonas [18, 19] propose a framework combining time-varying control barrier functions with STL to ensure satisfaction of complex tasks and cooperative behaviours among multiple agents. Similarly, Zhao and Yu [20] apply such methods to enable safe merging manoeuvres in autonomous vehicles.

1.2 Problem Description

We wish to solve the problem of untethered transportation of objects in microgravity. Given a limited set of free-flying agents and a high-level task specification, the agents must collaboratively transport a passive object to a desired state, defined in terms of target position and velocity, using rendezvous-preceded, gentle interactions. Unlike rigidly attached manipulation or impact-based methods, such interactions must operate under inherently under-actuated contact dynamics. The absence of physical coupling and full control authority makes it difficult to ensure both stability and precision, motivating the need for a structured planning and control framework that can explicitly account for these physical constraints while enabling safe and reliable transport.

1.2.1 Research Question

How can rendezvous-preceded interactions be specified within a planning and control framework to enable gentle, untethered collaborative transport of a free-floating, non-cooperative object in microgravity?

1.3 Purpose

The purpose of this thesis is to design and implement a software and control framework for untethered collaborative transport of passive objects in microgravity. To facilitate the long term viability of untethered transport, this thesis focuses on ensuring that transport is conducted through rendezvous preceded continuous interactions, thereby reducing the risk of damage associated with impulsive contacts. With continued development, this framework can support reliable autonomous space operations and enable more scalable and efficient transport solutions for satellite servicing, debris mitigation, and orbital assembly.

1.4 Goals

To achieve the stated purpose, the goals of this thesis include:

1. Developing a high-level planner capable of specifying rendezvous-preceded robot-object interactions and high-level task specification using STL and MIP.
2. Developing a control strategy that allows the robots to interact with the object in four distinct phases.
   (a) Rendezvous and Approach: Synchronize the velocity and position of the robots with the object and execute a controlled approach trajectory.
   (b) Untethered Contact: Establish safe contact with the object without mechanical grasping or interlocking, ensuring stability while minimizing risks of slippage or induced rotation.
   (c) Velocity Transfer: Impart a desired translational velocity to the object.
   (d) Separation and Stop: Disengage from the object, ensuring that the robots return to a stable state without imparting unintended forces or torques on the object.
3. Evaluate the developed methods in two-dimensional microgravity-analogue environments. This includes both physics-based simulations of robots on air bearings and experiments on a physical testbed with real robots operating on air-bearing platforms.

1.5 Research Methodology

The methodology used in this thesis largely follows the Design Science Research Process as presented in [21]:

1. Problem Identification and Motivation.
2. Defining the objectives of a solution.
3. Design and development.
4. Demonstration.
5. Evaluation.
6. Communication.

1.6 Delimitations

- No experiments will be conducted in actual or simulated three-dimensional microgravity. Instead, the evaluation will be restricted to two-dimensional analogue environments, where free-floating objects are supported by air-bearing platforms.
- Both robots and objects will be modelled with simplified circular geometries, and all interactions will be constrained to a 2D plane.
- The primary emphasis will be placed on simulation studies in 2D environments. Physical experiments on the air-bearing test-bed will only be pursued once sufficient progress has been achieved in simulation.
- Ground-truth motion data of robots and objects will be used for control and evaluation purposes, obtained either directly from simulation software or through motion-capture systems in experiments. On-board state estimation methods (e.g.perception) will not be developed, as this would significantly expand the scope of the thesis.
- The interaction framework will not consider highly irregular object geometries, deformable bodies, variable mass distributions, or matching of rotational velocities, as these add substantial modelling and control complexity that exceeds the scope of the project.

Chapter 2

Background

2.1 Related Work

Manipulation without rigid grasping has been widely studied in robotics under the umbrella of non-prehensile manipulation, where objects are moved through pushing, sliding, or rolling interactions rather than firm grasping. In [22], Wang *et al*. proposed a method leveraging reinforcement learning to enable dexterous manipulation of ungraspable objects, demonstrating accurate object repositioning, enabling either direct or environmental-assisted grasping. Similarly, to increase the adaptability of robotic manipulators used for on-orbit servicing, Li *et al*. [23] propose a method of mode-invariant trajectory optimization that enables manipulation of non-cooperative objects in space without a rigid grasp.

Closer to our approach are works on non-prehensile or untethered transportation techniques, where non-cooperative objects are moved and transported using collaborative multi-agent frameworks. In [24], a hybrid system integrates flow-matching co-generation and classical multi-robot motion planning to transport objects in a simulated terrestrial environment, highlighting the adaptability and potential of coordinated non-prehensile manipulation, albeit in structured terrestrial environments rather than microgravity. While these works highlight the potential of non-prehensile interaction and collaborative transport, they either focus on terrestrial environments or consider manipulation in space without addressing transport.

Building on these ideas, this thesis specifically extends the framework introduced by [14], who first demonstrated a method for collaborative transport in microgravity using impact-based interaction. Their formulation and methodological insights form the foundation for several components of the approach developed in this thesis. In particular, the parametrization of trajectories as Bézier curves and the encoding of STL specifications within MIP closely follows their presented method.

While the theoretical background and overall problem formulation are based on [14], this thesis advances the framework by extending it to utilize rendezvous-preceded interactions rather than direct impacts. This shift enables safer and less aggressive transport behaviours, which are more appropriate for delicate payloads and constrained operational environments. These extensions generalize the original framework and broaden its applicability, demonstrating how the underlying principles of collaborative transport can be maintained while significantly altering the interaction modality.

2.2 Signal Temporal Logic

STL [16] is widely used in cyber-physical and dynamical systems as a formal language for specifying real-time requirements such as safety constraints, temporal objectives, and conditional system behaviours. An STL formula is built from atomic predicates of the form

$$\mu := f(\mathbf{s}(t)) \geq 0,$$

where f is typically a linear function that evaluates some property of the real-valued signal $\mathbf{s}(t)$ at time t.

The syntax of STL is defined recursively as

$$\varphi ::= \mu \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \, \mathbf{U}_{[a,b]} \, \varphi_2,$$

where $\neg$ and $\wedge$ denote Boolean negation and conjunction, and $\mathbf{U}_{[a,b]}$ is the time-bounded *Until* operator. The formula $\varphi_1 \, \mathbf{U}_{[a,b]} \, \varphi_2$ states that φ_1 must hold continuously until φ_2 becomes true at some time within the interval $[a, b]$.

Using the *Until* operator, additional temporal operators can be defined. The *Eventually* operator $\mathbf{F}_{[a,b]} \, \varphi$ requires that φ holds at some time in the interval $[a, b]$, and the *Globally* operator $\mathbf{G}_{[a,b]} \, \varphi$ requires that φ holds at all times in $[a, b]$.

A summary of the Boolean semantics of STL is provided in (2.1).

$$
\begin{aligned}
(\mathbf{s}, t) \models \mu \;&\Leftrightarrow\; f(\mathbf{s}(t)) \geq 0 && \text{(2.1a)}\\
(\mathbf{s}, t) \models \neg\varphi \;&\Leftrightarrow\; (\mathbf{s}, t) \not\models \varphi && \text{(2.1b)}\\
(\mathbf{s}, t) \models \varphi_1 \wedge \varphi_2 \;&\Leftrightarrow\; (\mathbf{s}, t) \models \varphi_1 \text{ and } (\mathbf{s}, t) \models \varphi_2 && \text{(2.1c)}\\
(\mathbf{s}, t) \models \mathbf{F}_{[a,b]}\varphi \;&\Leftrightarrow\; \exists t' \in [t+a, t+b] : (\mathbf{s}, t') \models \varphi && \text{(2.1d)}\\
(\mathbf{s}, t) \models \mathbf{G}_{[a,b]}\varphi \;&\Leftrightarrow\; \forall t' \in [t+a, t+b] : (\mathbf{s}, t') \models \varphi && \text{(2.1e)}\\
(\mathbf{s}, t) \models \varphi_1 \mathbf{U}_{[a,b]}\varphi_2 \;&\Leftrightarrow\; \exists t' \in [t+a, t+b] : (\mathbf{s}, t') \models \varphi_2 && \text{(2.1f)}\\
&\quad\;\; \wedge\; \forall \tau \in [t, t'] : (\mathbf{s}, \tau) \models \varphi_1
\end{aligned}
$$

To support optimization-based testing and trajectory synthesis, STL can also be equipped with a *robustness* value $\rho(\mathbf{s}, t, \varphi)$, which quantifies not only whether a signal satisfies a formula but also by what margin. Robust semantics assign a positive value when the formula is satisfied, a negative value when it is violated, and a magnitude that reflects the distance to the satisfaction boundary. Equations (2.2a)–(2.2d) illustrate some representative robustness measures. For a complete treatment of STL semantics we refer the reader to [16, 25].

$$
\begin{aligned}
\rho(\mathbf{s}, t, \varphi) > 0 \;&\Leftrightarrow\; (\mathbf{s}, t) \models \varphi && \text{(2.2a)}\\
\rho(\mathbf{s}, t, \varphi) < 0 \;&\Leftrightarrow\; (\mathbf{s}, t) \not\models \varphi && \text{(2.2b)}\\
\rho(\mathbf{s}, t, \mathbf{G}_{[a,b]}\,\varphi) &= \min_{t' \in [t+a, t+b]} \rho(\mathbf{s}, t', \varphi) && \text{(2.2c)}\\
\rho(\mathbf{s}, t, \mathbf{F}_{[a,b]}\,\varphi) &= \max_{t' \in [t+a, t+b]} \rho(\mathbf{s}, t', \varphi) && \text{(2.2d)}
\end{aligned}
$$

2.3 Mixed Integer Programming

MIP is an optimization framework commonly used in decision making, planning, and control. It is particularly powerful due to its ability to combine a *mixture* of continuous and binary decision variables, within a single model. This is especially useful for dynamic systems that feature both continuous quantities (e.g., positions, velocities, energies) and discrete decisions (e.g., on/off, left/right, activate/deactivate). By combining these two types of variables, MIP facilitates the synthesis of plans, trajectories, and control strategies that simultaneously satisfy kinematic, physical, and logical constraints while still pursuing a specified objective.

In the context of robotics and control, a MIP model is typically composed of decision variables that indicate the state of a system, and constraints that encode system dynamics, kinematics, safety requirements, and logical relations. Moreover, the model includes an objective function then guides the optimizer toward a desirable solution, such as minimizing time, maximizing safety margins, or achieving a goal state efficiently. The exact structure of the model determines the terminology used: if all constraints and the objective are linear, the model is referred to as *Mixed Integer Linear Programming (MILP)*; if quadratic terms are used in the objective, it becomes *Mixed Integer Convex Programming (MICP)*, and so on.

For the purposes of this thesis, a deep mathematical treatment of MIP is not required. The underlying theory is well established in the literature and is therefore only introduced to the extent necessary for the present work. Interested readers are referred to [26–28] for a more comprehensive background. A practical challenge when working with MICP models is that even simple logical constraints can become notationally cumbersome when written explicitly. To avoid interrupting the flow of later chapters with low-level formulations, this background section introduces the standard modelling primitives that will be used throughout the thesis. The examples below illustrate how logical relations and hybrid behaviour can be represented using linear constraints and mixed-integer decision variables.

Polyhedral Region Membership

Atomic conditions such as remaining inside a safe region or satisfying a state-dependent predicate are represented as halfspace constraints of the form

$$a^\top x_t \le b.$$

Logical Implication via Big-M

Binary variables enable the encoding of logical implications. A statement of the form

$$z_t = 1 \;\Rightarrow\; a^\top x_t \leq b$$

is enforced using the standard Big-M relaxation:

$$a^\top x_t \leq b + M(1 - z_t),$$

where M is chosen large enough so that the constraint becomes inactive when $z_t = 0$. Such constructions form the backbone of MICP-based encodings of Boolean STL operators.

Disjunction (OR) Constraints

To represent a logical disjunction such as

$$a^\top x_t \leq b \quad \text{or} \quad c^\top x_t \leq d,$$

we introduce two binary variables $z_{1,t}, z_{2,t}$ and enforce

$$\begin{aligned} z_{1,t} + z_{2,t} &= 1, \\ a^\top x_t &\leq b + M(1 - z_{1,t}), \\ c^\top x_t &\leq d + M(1 - z_{2,t}). \end{aligned}$$

At most one constraint must hold tightly, ensuring that at least one of the corresponding regions is selected.

Binary Activation of Equality Constraints

Sometimes a binary variable must enforce that a particular equality holds *only when activated*. For a constraint of the form

$$z_t = 1 \;\Rightarrow\; x_t = c,$$

we use both upper and lower Big-M bounds:

$$\begin{aligned} x_t &\leq c + M(1 - z_t), \\ x_t &\geq c - M(1 - z_t). \end{aligned}$$

When $z_t = 1$, both inequalities reduce to $x_t = c$. When $z_t = 0$, the bounds become inactive and x_t is unconstrained within the range implied by M.

2.4 Bézier Curves

Bézier curves are a family of parametric curves defined by polynomial functions and a finite set of control points. They are widely used in computer graphics and geometric modelling and recently also gained increasing attention in motion planning [29]. Their polynomial structure makes them particularly suitable for trajectory generation [30, 31], as it enables a compact way of controlling and enforcing non-linear trajectories.

A Bézier curve of degree n is defined by a sequence of $n+1$ control points

$$\mathbf{P}_0, \mathbf{P}_1, \dots, \mathbf{P}_n \in \mathbb{R}^d,$$

where d is the dimension of the space (typically $d = 2$ or 3). The *degree* of the curve determines the order of the underlying polynomial, which in turn affects the curves complexity or flexibility. Higher-degree curves allow for more complex shapes, but they also require more control points and computational effort.

A Bézier curve is parameterized over the phasing parameter $s \in [0, 1]$ and expressed as a weighted sum of its control points using the Bernstein basis polynomials:

$$\mathbf{B}(s) = \sum_{i=0}^{n} \mathbf{P}_i \, b_{i,n}(s),$$

where the Bernstein polynomials of degree n are given by

$$b_{i,n}(s) = \binom{n}{i} (1 - s)^{n-i} s^i.$$

The control points act as geometric handles that define the shape of the curve. Although a Bézier curve does not, in general, interpolate all of its control points, it is guaranteed to lie entirely within the convex hull defined by them. This property is especially valuable in trajectory planning, where geometric feasibility guarantees are essential. An example of a Bézier curve is shown in Fig. 2.1(a), where the control points form a convex hull enclosing the curve. Notice that the first and last control points determine the start and end of the curve. Furthermore, the derivative of a Bézier curve is itself a Bézier curve of one degree lower, whose control points are given by linear combinations of the original ones:

$$\frac{d\mathbf{B}(s)}{ds} = n \sum_{i=0}^{n-1} (\mathbf{P}_{i+1} - \mathbf{P}_i)\, b_{i,n-1}(s),$$

which shows that the derivative curve is defined by the n control points

$$\mathbf{P}'_i = n(\mathbf{P}_{i+1} - \mathbf{P}_i), \qquad i = 0, \ldots, n-1.$$

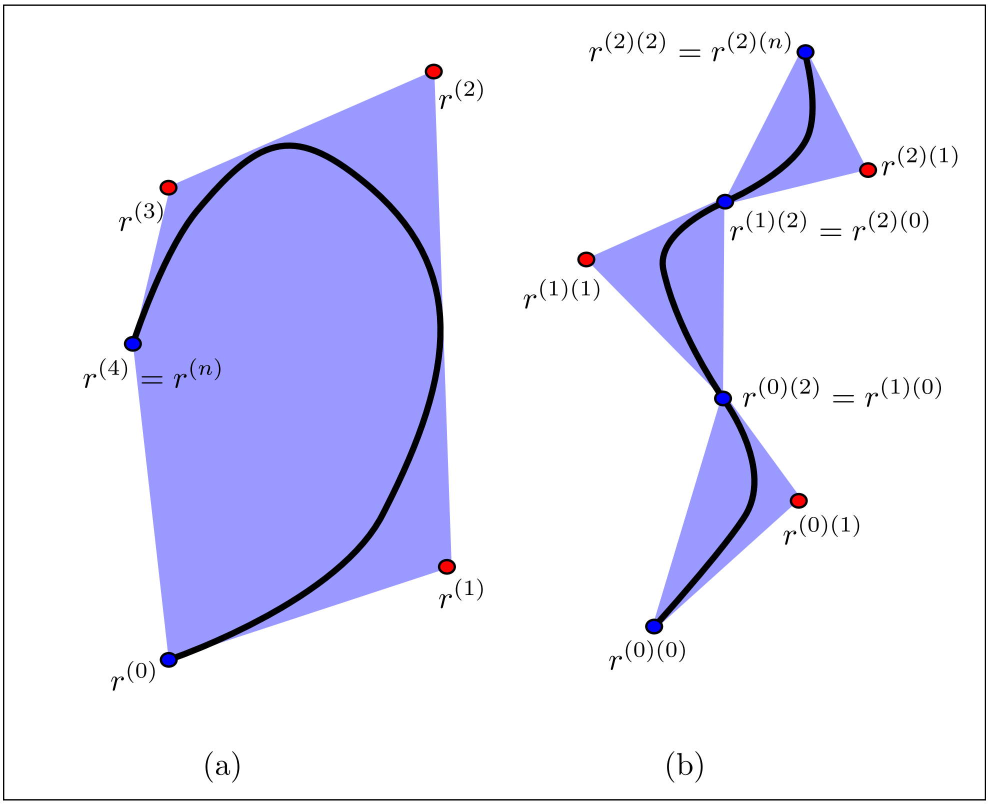

Figure 2.1: (a) A Bézier curve showing the start and end points (blue dots), control points (red dots), and the associated convex hull (shaded region). (b) A robot trajectory parameterized by multiple Bézier curves connected at their endpoints.

2.4.1 Trajectory Parametrization

Parametrizing physical trajectories using Bézier curves, as illustrated in Fig. 2.1(b), offers significant advantages over fine-grained discretization, as it requires fewer parameters to represent smooth trajectories. Reducing the number of parameters directly decreases the computational complexity of motion planning and trajectory optimization problems, a particularly valuable property in the presence of integer variables. While a purely spatial parametrization can be achieved using a single Bézier curve, such an approach ignores the temporal aspect of a trajectory, which is critical for satisfying temporal constraints.

To address this, [30] proposes a method for parametrizing trajectories over time using two Bézier curves: a *spatial* curve $r(s)$ and a *temporal* curve $h(s)$, both defined over the phasing parameter $s \in [0, 1]$. The actual physical trajectory $p(t)$ is then expressed by composing these two curves. Specifically, the trajectory evaluated at s is given by

$$r(s) = p(h(s)), \tag{2.3}$$

and, by the chain rule, its derivative satisfies

$$\dot{r}(s) = \dot{p}(h(s))\,\dot{h}(s).$$

This parametrization separates the geometric shape of the trajectory from its temporal evolution, enabling independent control of spatial and timing characteristics. Since these curves may span an arbitrary distance or time, this approach offers increased efficiency compared to fine-grained discretization, and linking multiple curves together enables a compact representation of complex trajectories.

Chapter 3

Method

This chapter presents the framework used to address the problem of collaborative untethered transportation in microgravity. The overall structure and several key components are adapted from the approach presented in [14]. In particular, the hierarchical structure of the framework and the approach taken in Sections 3.2, 3.3.1, 3.3.2, 3.3.3.4 and 3.3.3.5 are adapted from their work. We first formally state the problem and introduce the proposed framework before describing each component of the method in detail.

3.1 Problem Formulation

Consider the set of robots $\mathcal{R} = \{R_1, \dots, R_{|\mathcal{R}|}\}$ and the set of objects $\mathcal{O} = \{O_1, \dots, O_{|\mathcal{O}|}\}$, and define $\mathcal{S} = \mathcal{R} \cup \mathcal{O}$ as the set of all systems involved in the task. Each system $S \in \mathcal{S}$ is described by the state vector $x_S = [\,p_S,\ q_S,\ \dot{p}_S,\ \dot{q}_S\,]$, where p_S and q_S denote its position and orientation.

Because the solution must explicitly determine *when* and *between whom* interactions occur, we introduce the binary interaction variable $\mathit{interact}(S_i, S_j, t) \in \{\perp, \top\}$, which indicates whether system S_i is interacting with system S_j at time t.

We can now formulate the problem for collaborative transport using rendezvous preceded interactions as follows:

$$\min_{x_{R_i}, u_{R_i}, x_{O_j}} \quad J(\mathbf{x}_{\mathcal{R}}, \mathbf{u}_{\mathcal{R}}, \mathbf{x}_{\mathcal{O}}), \tag{3.1a}$$

subject to

$$[\mathbf{x}_{\mathcal{R}}, \mathbf{x}_{\mathcal{O}}] \vDash \varphi, \tag{3.1b}$$

$$\dot{\mathbf{x}}_R \in \mathcal{V}_{\mathcal{R}}, \; \dot{\mathbf{x}}_O \in \mathcal{V}_{\mathcal{O}}, \tag{3.1c}$$

$$\dot{x}_{R_i} = \left\{ \begin{bmatrix} \dot{p}_{R_i} & \dot{q}_{R_i} & u_{R_i} \end{bmatrix}^T \right. \tag{3.1d}$$

$$\dot{x}_{O_i} = \begin{cases} \begin{bmatrix} \dot{p}_{R_j} & \dot{q}_{R_j} & 0 \end{bmatrix}^T & \Leftarrow \mathit{interact}(O_i, R_j, t) \\ \begin{bmatrix} \dot{p}_{O_i} & \dot{q}_{O_i} & 0 \end{bmatrix}^T & \text{otherwise} \end{cases} \tag{3.1e}$$

$$\begin{aligned} &\forall\, i, j \in \mathcal{S},\; i \neq j,\; \forall\, t \in [t_0, t_f] : \\ \neg \mathit{interact}(S_i, S_j, t) \Rightarrow \|p_{S_i}(t) - p_{S_j}(t)\|_2 &> rad_{S_i} + rad_{S_j} \end{aligned} \tag{3.1f}$$

$$\begin{aligned} &\forall\, i, j \in \mathcal{S},\; i \neq j,\; \forall\, t \in [t_0, t_f] : \\ \mathit{interact}(S_i, S_j, t) \Rightarrow \|p_{S_i}(t) - p_{S_j}(t)\|_2 &= rad_{S_i} + rad_{S_j} \end{aligned} \tag{3.1g}$$

where $\mathbf{x}_{\mathcal{R}}$ and $\mathbf{x}_{\mathcal{O}}$ denote the aggregated states of all robots and objects, respectively, and $\mathcal{V}_{\mathcal{R}}$ and $\mathcal{V}_{\mathcal{O}}$ represent the permissible velocity bounds for each group. The state of an individual system $S_i \in \{R, O\}$ is given by $x_{S_i} = [p, q, \dot{p}, \dot{q}]$, u_{R_i} denotes the control input applied to robot R_i and rad its radius. The term φ captures the STL specifications, including workspace limits, transportation objectives, collision avoidance, and obstacle avoidance.

Equations (3.1b)–(3.1c) ensure that all system trajectories satisfy the high-level STL specifications and remain within the allowable velocity bounds. Equations (3.1d)–(3.1e) enforce that each robot's state is governed by its own input, while objects adopt the state of the interacting robot during an interaction. Equation (3.1f) ensures that the distance between any two non-interacting systems remains larger than the sum of their radii. Finally, (3.1g) specifies the required conditions for the transfer of normal forces between two systems during an interaction.

3.2 Approach

Following the hierarchical framework presented in [14], our method is divided into three main components, as illustrated in Fig. 3.1. First, an offline planner operates on simplified kinematic models, treating all robots and objects as point masses. This planner generates nominal trajectories and determines when, where, and with whom robots and objects should interact. Second, an online re-planner incorporates the physical dimensions of all systems and enforces feasibility constraints on trajectories and interactions, while continuously updating plans based on the current estimated states of all robots and objects. Finally, a Model Predictive Control (MPC) layer ensures accurate tracking of the generated trajectories and the reliable execution of the planned interactions.

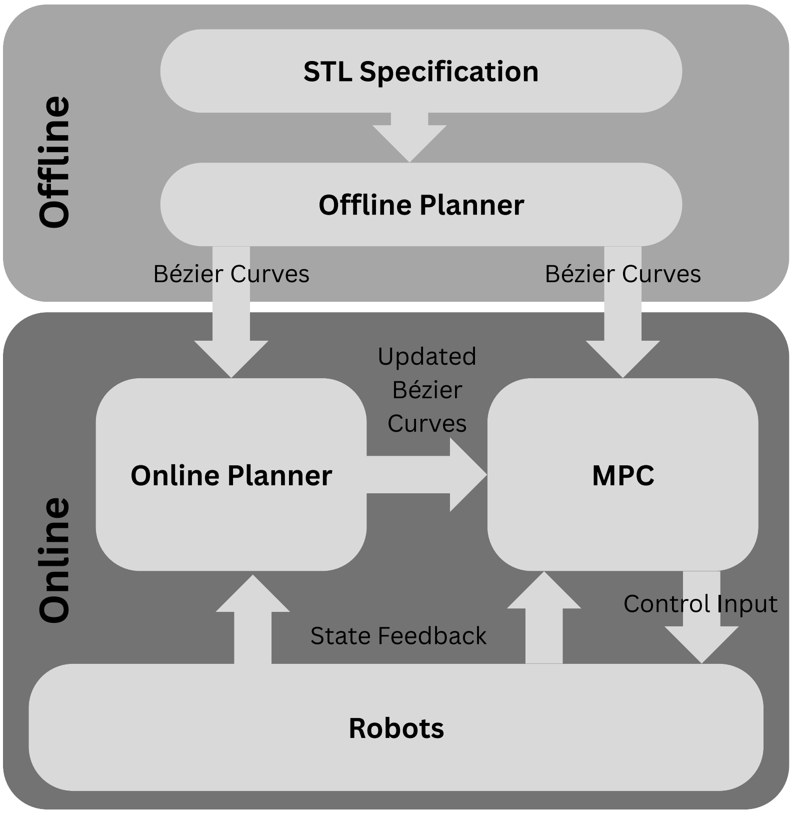

Figure 3.1: High-level overview of the proposed hierarchical planning and control framework, illustrating the interaction between offline planning, online re-planning, and low-level model predictive control.

3.3 Offline Planner

3.3.1 Introduction

For the synthesis of a plan, it is necessary to determine how the collaborative transport task should be executed in order to best satisfy the overall objective. At the same time, the generated motions must resemble physically feasible trajectories and interactions as closely as possible. To meet these requirements, we employ a MICP planner. Due to the computational complexity associated with solving such problems, the MICP planning is performed offline. This planner supplies the online planner and the MPC with the nominal plans (as seen in Fig. 3.1), in the form of Bézier curves that serve as a reference for subsequent re-planning and feedback control.

To further reduce computational demands, both robots and objects are modelled as point masses, and their orientations and rotational velocities are omitted from the planning formulation. Additionally, directly reasoning over continuous trajectory representations in the workspace would severely hinder MICP performance, as it would require parametrizing robot trajectories with an intractably large number of variables. To address this, and following the approach in [14], detailed in Section 2.4.1, we parametrize robot and object trajectories using Bézier curves. This representation significantly reduces the number of decision variables required for the planner to operate, enabling efficient computation of feasible trajectories. As demonstrated in [32], this approach provides substantial computational benefits while preserving trajectory expressiveness.

Since objects are not actuated and cannot change their state autonomously, their motion is only permitted to change through interactions with robots. Consequently, object trajectories are restricted to first-degree Bézier curves, which correspond to constant-velocity motion between interactions. In contrast, robots are actuated and capable of changing their motion, and their trajectories are therefore parameterized by second-degree Bézier curves or higher.

3.3.2 MICP Encoding of STL Specifications

Using the modelling primitives introduced in Section 2.3, STL formulas can be encoded exactly as mixed-integer linear constraints. For example, a predicate with the form

$$\mu(x_t) \equiv a^\top x_t \leq b.$$

can be encoded using a binary variable $z_{\mu,t}$ as follows:

$$a^\top x_t \leq b + M(1 - z_{\mu,t}),$$
$$a^\top x_t \geq b - M z_{\mu,t}.$$

In general, each STL subformula φ is associated with a binary variable $z_{\varphi,t} \in \{0, 1\}$ that denotes whether the formula is satisfied at time t. Compositional STL operators (e.g., conjunctions, disjunctions, and other temporal operators) are encoded by linking these binary variables through standard logical and temporal constructions. For details the reader is referred to [32, 33].

3.3.2.1 STL Integration into the MICP Objective

As discussed in Section 2.2, each atomic predicate is assigned a robustness value (e.g., space-based or time-based). Following the approach in [14, 33], these robustness values are propagated through the STL syntax tree via min and max operators. The resulting robustness at the root formula φ provides a scalar measure of spatial/temporal satisfaction. This root robustness value is then incorporated directly into the MICP as part of the objective function. We can thus formulate the spatial-robust planning problem as follows:

$$\max_{r_{R_i}, h_{R_i}, r_{O_j}, h_{O_j}} \quad J(r_{R_i}, h_{R_i}, r_{O_j}, h_{O_j}) = \rho_\varphi(\mathbf{x}_\mathcal{R}, \mathbf{x}_\mathcal{O}), \tag{3.2a}$$

s.t.

$$\dot{x}_{R_i} = \left\{ \begin{bmatrix} \dot{p}_{R_i} & u_{R_i} \end{bmatrix}^T \right. \tag{3.2b}$$

$$\dot{x}_{O_i} = \begin{cases} \begin{bmatrix} \dot{p}_{R_j} & 0 \end{bmatrix}^T & \Leftarrow \textit{interact}(O_i, R_j, t) \\ \begin{bmatrix} \dot{p}_{O_i} & 0 \end{bmatrix}^T & \text{otherwise} \end{cases} \tag{3.2c}$$

$$\forall\, i, j \in \mathcal{S},\ i \neq j,\ \forall\, t \in [t_0, t_f] : \tag{3.2d}$$
$$\textit{interact}(S_i, S_j, t) \Rightarrow \|p_{S_i}(t) - p_{S_j}(t)\|_2 = 0$$
$$\text{Eqs. (3.1b), (3.1c), (3.1f)}$$

3.3.3 MICP encoding of kinematic and conditional constraints

This section presents the kinematic and conditional constraints used to encode trajectories and interactions within the MICP. For clarity and readability, we omit the full MICP formulations and instead provide simplified representations of the constraints. Nevertheless, all constraints discussed here are implemented using the standard primitives introduced in Section 2.3. To support a consistent and interpretable presentation, the notation used throughout this chapter is summarized in Table 3.1. Note that, due to the chosen trajectory parametrization, velocity cannot be constrained directly. Instead, velocity constraints are enforced through linear constraints on the derivatives of the spatial ($\dot{r}$) and temporal ($\dot{h}$) curves separately. Moreover, although interactions may occur at arbitrary continuous times, we restrict them to occur only between a single robot curve and a single object curve at any given moment. Interactions are further limited to take place only at the start or end of these curves. To condition the activation of constraints on whether an interaction occurs, we introduce binary variables $z_{R,O} \in \mathbb{B}^{(N_R \times N_O)}$, where N_R and N_O denote the number of Bézier curves assigned to robot R and object O, respectively. These binaries serve a role analogous to *interact*(O_i, R_j, t), but are defined per curve pair rather than per discrete time step t.

Example 1. *Consider the example transport scenario shown in Fig. 3.2(a). The objective is to transport an object to a designated goal region through untethered, continuous force transfer between multiple robots and the object. Figure 3.2(b) illustrates representative trajectories highlighting the sequence of phases required to accomplish this transport. Specifically, the planner must determine when and where interactions occur, as well as which robot engages the object at each stage, in order to successfully facilitate the overall motion. Each phase of the transport is subject to distinct constraints, depending on whether the task requires rendezvous or force transfer. Robot R_1 first performs an approach and rendezvous with a stationary object (A_1), after which it applies a positive force along the y-axis to accelerate the object toward the goal region (I_1). In contrast, robot R_2 must rendezvous with a moving object (A_2), before applying a decelerating force to bring the object to rest within the goal region (I_2).*

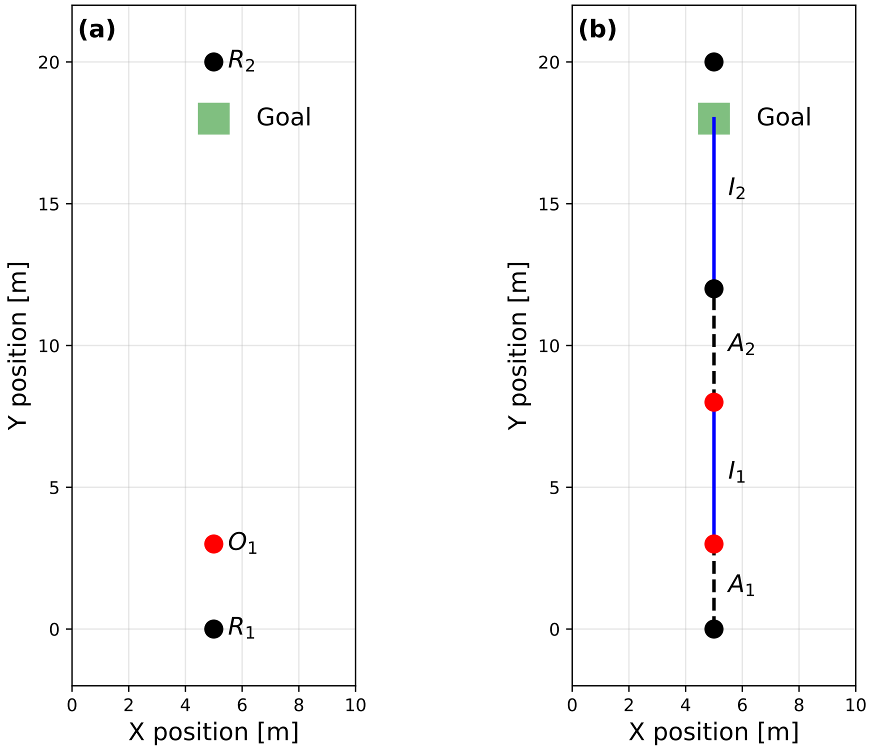

Figure 3.2: Example transport scenario showing (a) the problem setup, consisting of two robots (R_1 and R_2) transporting an object (O_1) to a goal region, indicated in green. (b) Example trajectories highlighting the distinct phases of transport, namely approach and rendezvous (A_1 and A_2), followed by interaction phases (I_1 and I_2), during which continuous acceleration is imparted to the object.

3.3.3.1 Continuity Constraints

To ensure that the generated trajectories are kinematically feasible and do not include discrete jumps between Bézier segments we enforce the following constraints for robots:

$$\begin{aligned} h_R^{(k)(n)} &= h_R^{(k+1)(0)} \\ r_R^{(k)(n)} &= r_R^{(k+1)(0)} \\ \dot{h}_R^{(k)(n)} &= \dot{h}_R^{(k+1)(0)} \\ \dot{r}_R^{(k)(n)} &= \dot{r}_R^{(k+1)(0)} \end{aligned} \qquad \forall k \in \mathbb{N}_{[0,N_R-2]} \tag{3.3}$$

Similarly, we enforce temporal continuity for all object Bézier curves

r	Spatial curve
h	Temporal curve (time)
$\dot{r}$	Derivative of spatial curve
r_R	Spatial curve of robot R
r_O	Spatial curve of object O
$r^{(k)}$	Curve with index k
$r^{(k)(0)}$	First control point of curve k
$r^{(k)(n)}$	Last control point of curve k
$\hat{r}$	Re-planned curve
P_R^t	Measured state of robot R at time t (position $[x, y]$)
ρ_R	Radius of robot R
N_R	Number of Bézier curves assigned to robot R
$z_{(R,O)}[k, j]$	Interaction binary for curve k of robot R and curve j of object O
m_R	Mass of robot R

Table 3.1: Notation

$$h_O^{(j)(n)} = h_O^{(j+1)(0)} \qquad \forall j \in \mathbb{N}_{[0,N_O-2]}, \tag{3.4}$$

whereas position and velocity continuity is only enforced for non-interacting curves as follows:

$$\begin{aligned} r_O^{(j)(n)} &= r_O^{(j+1)(0)} \\ \dot{h}_O^{(j)(n)} &= \dot{h}_O^{(j+1)(0)} \\ \dot{r}_O^{(j)(n)} &= \dot{r}_O^{(j+1)(0)} \end{aligned} \quad \Leftarrow \quad \begin{aligned} &z_{(R,O)}[k, j] = \perp \\ &\forall R \in \mathcal{R} \ \ \forall k \in \mathbb{N}_{[0,N_R-1]} \end{aligned} \quad \forall j \in \mathbb{N}_{[0,N_O-2]} \tag{3.5}$$

This distinction arises from the difference in trajectory parameterization between robots and objects. Object trajectories are restricted to first-degree Bézier curves, which can only represent constant-velocity motion. As a result, objects cannot represent changes in velocity or non-linear motion within a single curve segment. Consequently, when an interaction occurs, the updated object state cannot be represented continuously within the same curve. To address this, we therefore permit discontinuities in position and velocity between an interacting curve and its *succeeding* curve, as illustrated in Fig. 3.3.

3.3.3.2 Interaction Constraints

For a robot and an object to engage, the interaction must be preceded by a rendezvous, meaning that their positions and velocities coincide at a specific point in time. Accordingly, we define an interaction between robot Bézier segment k and object Bézier segment j as follows:

$$\begin{aligned} h_R^{(k)(0)} &= h_O^{(j)(0)} \\ r_R^{(k)(0)} &= r_O^{(j)(0)} \\ \dot{h}_R^{(k)(0)} &= \dot{h}_O^{(j)(0)} \\ \dot{r}_R^{(k)(0)} &= \dot{r}_O^{(j)(0)} \end{aligned} \quad \Longleftarrow \quad z_{(R,O)}[k,j] = \top \tag{3.6}$$

During an interaction, the robot and object states are conceptually assumed to be identical, such that any change in the robots state is also applied to the object. As discussed in Section 3.3.3.1, object trajectories cannot capture velocity changes within a single curve segment. Instead, discrete updates to the object state are allowed after an interaction (see (3.5)), and the object state resulting from the interaction is applied at the start of the subsequent trajectory segment:

$$\begin{aligned} h_O^{(j+1)(0)} &= h_R^{(k+1)(0)} \\ r_O^{(j+1)(0)} &= r_R^{(k+1)(0)} \\ \dot{h}_O^{(j+1)(0)} &= \dot{h}_R^{(k+1)(0)} \\ \dot{r}_O^{(j+1)(0)} &= \dot{r}_R^{(k+1)(0)} \end{aligned} \quad \Longleftarrow \quad z_{(R,O)}[k,j] = \top \tag{3.7}$$

As an example, Fig. 3.3 illustrates the discontinuity in the object state following an interaction. When the object is interacted with from rest, its velocity (and position) remains unchanged during the interaction and is only updated to match the robot state at the start of the *next* trajectory segment.

For notational clarity, throughout the remainder of this paper we refer to the object state at the end of the interaction as $r_O^{(j)(n)}$, even though the actual state update occurs at the beginning of the subsequent curve.

3.3.3.3 Robot Interaction Constraints

Since interactions are required to be *untethered*, no mechanical coupling exists between the robot and the object. As illustrated in Fig. 1.1(c), this limits the forces that can be transferred from the robot to the object to positive normal forces (pushing) only. Consequently, the synthesized robot plans cannot

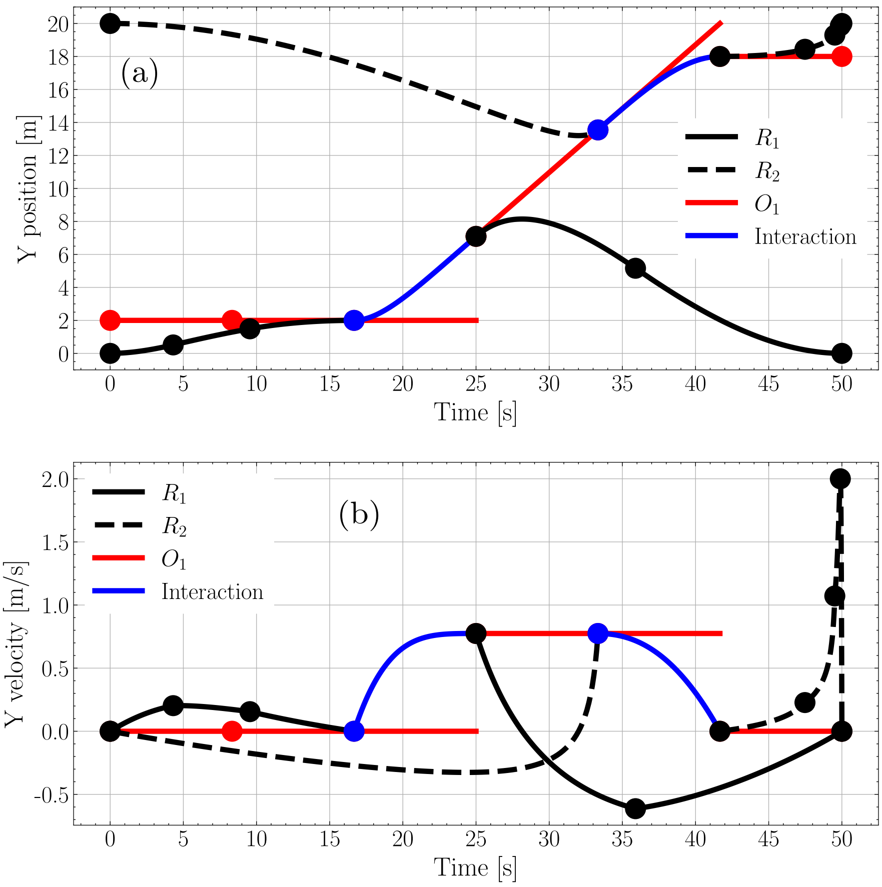

Figure 3.3: Trajectory generated by the offline planner for the example scenario in Fig. 3.2, illustrating a discontinuity in the object position (a) and velocity (b) between an interaction and its succeeding curve. During the interaction, the object velocity is held constant, and the state update is applied at the start of the subsequent curve. Dots mark the start of a new curve.

apply force in an arbitrary direction during an interaction. Instead, for each interaction, the planner must commit to a specific quadrant of allowable force directions for the entirety of the interacting curve. This follows from the fact that a change in the sign of the applied force would require a change in contact configuration. To capture this limitation, the planner selects the sign of the applied force in each dimension. The binary variables F^+ and F^- indicate the chosen sign of the acceleration (or force) along a given coordinate axis, for the entirety of an interacting curve. *

$$\begin{aligned}
&\dot{r}_R^{(k)(i+1)} - \dot{r}_R^{(k)(i)} \geq 0 \quad \Leftarrow F^+ \\
&\dot{r}_R^{(k)(i+1)} - \dot{r}_R^{(k)(i)} \leq 0 \quad \Leftarrow F^- \\
&\dot{h}_R^{(k)(i+1)} - \dot{h}_R^{(k)(i)} = 0 \\
&\forall i \in \mathbb{N}_{[0,n-1]} \quad \Longleftarrow \\
&z_{(R,O)}[k,j] = \top \ \exists j \in \mathbb{N}_{[0,N_O-1]} \ \exists O \in \mathcal{O}
\end{aligned} \tag{3.8}$$

Example 2. *Consider again the transport scenario shown in Fig. 3.2(a). If left unconstrained, the planner could generate a single interaction phase that replaces phases I_1 and I_2, in which robot R_1 both accelerates and subsequently decelerates the object within one continuous interaction.*

From the planner's perspective, such a solution is admissible, as the robot and object are modelled as point masses and are effectively treated as being rigidly coupled during an interaction. However, in a physical setting, this manoeuvrer would require the robot to disengage from the object, move around it, and re-establish contact from the opposite side before applying a force in the opposite direction. Consequently, the transport would necessarily involve two distinct interaction phases rather than a single continuous one.

To prevent such physically infeasible solutions, constraint (3.8) restricts the force applied during an interaction to a single sign along each coordinate axis. At the same time, the constraint allows a combination of forces across different axes, enabling diagonal pushing motions if required.

3.3.3.4 Workspace Constraints

Following the method in [14], containment of a Bézier curve within a Region of Interest (RoI) can be enforced by constraining all of its control points to satisfy the linear inequalities defining the RoI polytope. Let the RoI be given in the half-space representation $Hx \leq b$, and let n^{faces} denote the number of faces of the polytope. Then the condition

*Because $v = \frac{\dot{r}}{\dot{h}}$, encoding a constant force direction directly in the MICP would introduce a quadratic constraint, resulting in a Mixed-Integer Quadratically Constrained Program (MIQCP) with substantially higher computational cost. To avoid this, we instead enforce a constant value for $\dot{h}$ in constraint (3.8), and apply the force constraint only to $\dot{r}$, sacrificing a small amount of flexibility for a significant gain in efficiency.

$$x_S \in \text{RoI} \iff \left(\bigwedge_{f=1}^{n^{\text{faces}}} \bigwedge_{i=0}^{n} \left(H^f r^{(j)(i)} \leq b^f \right) \right), \quad \forall j \in \mathbb{N}_{[0,N_S-1]} \tag{3.9}$$

ensures that every control point i of the j-th Bézier segment lies inside all bounding half-spaces of the RoI.

Conversely, collision avoidance with a RoI can be imposed by enforcing that at least one control point violates one of the supporting half-spaces. This can be expressed as

$$x_S \notin \text{RoI} \iff \left(\bigvee_{f=1}^{n^{\text{faces}}} \bigwedge_{i=0}^{n} \left(H^f r^{(j)(i)} \geq b^f \right) \right), \quad \forall j \in \mathbb{N}_{[0,N_S-1]} \tag{3.10}$$

where the disjunction over faces ensures that each Bézier segment remains strictly outside at least one face of the polytope.

3.3.3.5 Acceleration Attenuation

Since the offline planner determines the full kinematic and interaction structure of the control stack without considering the dynamics, ensuring feasible trajectories with bounded accelerations for both robots and objects is a key design consideration. However, due to the chosen trajectory parametrization, directly constraining accelerations is not possible. The trajectory is defined by two sets of curves, and taking the second derivative of (2.3) results in non-convex expressions, rendering them unsuitable for inclusion in a MICP formulation.

To address this limitation, accelerations are instead attenuated in the cost function, following the approach in [14, 30], by introducing a quadratic penalty term

$$J\left(r_{S_i}, h_{S_i}\right) = \sum_{k=0}^{N-1} \ddot{r}_{S_i}(k)^\top Q_{\ddot{r}} \ddot{r}_{S_i}(k) + \ddot{h}_{S_i}(k)^\top Q_{\ddot{h}} \ddot{h}_{S_i}(k), \tag{3.11}$$

which is added to the cost function in (3.2a).

In addition to the terms discussed above, the offline planning problem includes further constraints detailed in Appendix A.

3.4 Online re-planner

3.4.1 Introduction

The division of the planning architecture into two distinct modules is motivated by two fundamental limitations of the offline planner. First, due to the complexity of the optimization problem and the long planning horizon, the offline planner cannot be executed reliably in real time. Second, the synthesized offline plans do not account for real-time observations of robot and object states, nor their physical dimensions. To address these limitations, and following the methodology in [14], an online re-planner is introduced.

Since its primary purpose is reliable on-line execution, the re-planner is deliberately limited in scope. It neither re-evaluates global STL satisfaction nor reconstructs the obstacle and collision-avoidance structure encoded in the offline solution. Instead, its role is to locally adjust the *preceding* and *interaction* Bézier curves of a *robot* as the time of a planned interaction approaches. These adjustments account for updated robot and object states, their physical dimensions, and ensure that the robot is positioned such that the resulting normal force transfer induces the desired change in object velocity. Following the challenges discussed in Section 3.3.3.5, the re-planner attenuates robot accelerations during these two segments, rather than enforcing explicit acceleration constraints.

Example 3. *Consider again the example transport problem, now accounting for the physical dimensions of the robots and object, as well as online adaptation to updated robot and object states, as illustrated in Fig. 3.4. As the planned trajectories are adjusted to account for physical size, the point-mass assumption used in the offline planner is no longer valid and exact consensus of robot and object position during rendezvous is no longer feasible. Instead, rendezvous must be enforced such that the robot and object establish contact at the boundaries of their respective hulls.*

Furthermore, the assumption that the robot and object are rigidly coupled during interaction is relaxed. Changes in the robot state are no longer instantaneously transferred to the object independent of relative positioning. As a result, rendezvous must be achieved with an additional alignment constraint, ensuring that the robot is positioned along the intended direction of force prior to interaction. For example, during interaction I_1*, robot* R_1 *must rendezvous with object* O_1 *on the side opposing the desired direction of motion, such that the applied contact force accelerates the object along the positive* y*-axis. Conversely, during interaction* I_2*, robot* R_2 *must approach the object*

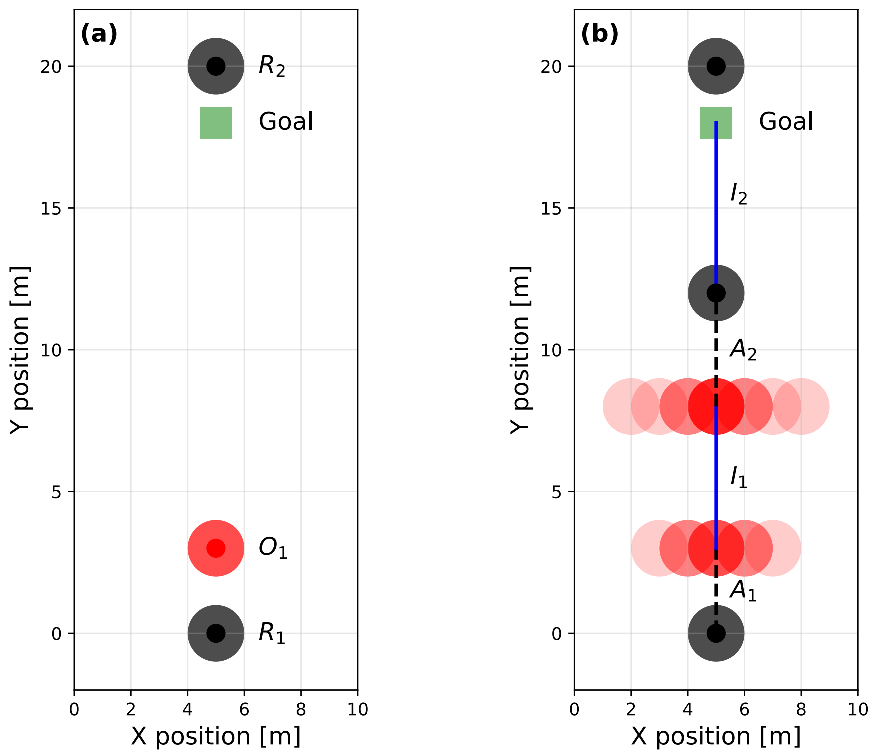

Figure 3.4: Example transport scenario with physical dimensions accounted for, showing (a) the problem setup, consisting of two robots (R_1 and R_2) transporting an object (O_1) to a goal region, indicated in green. (b) Example trajectories illustrating the distinct phases of transport and the uncertainty in object position arising between interaction phases.

from the opposite side, enabling the application of a force that decelerates the object and brings it to rest within the goal region.

Finally, in a real world setting, the object state between interaction phases is subject to significant uncertainty. Execution errors during interaction I_1 may cause the object to deviate substantially from its planned trajectory, necessitating re-planning of the subsequent approach and interaction phases, A_2 and I_2, to ensure successful task completion.

To support interpretability of the re-planner, the planning process is formulated around three discrete time instants, as illustrated in Fig. 3.5. The time $T = t$ denotes the instant at which re-planning is triggered, when updated measurements of robot and object states are obtained and the re-planning problem is solved. Time instants t_S and t_F correspond to the start and end of the interaction Bézier curve, respectively. In the following sections, decision variables are denoted by a hat $\widehat{(\cdot)}$, while all remaining variables are treated as known quantities obtained from measurements or from the nominal plans.

3.4.2 Re-planning Trigger

Re-planning is triggered prior to an upcoming interaction between a robot and an object. Let the Bézier curve on which this interaction occurs be indexed by I. The *preceding* robot curve, indexed by $I-1$, is adjusted to ensure the robot rendezvous with the object based on the latest observations. Its initial condition is defined using the re-planning trigger time t and the measured robot state:

$$\hat{h}_R^{(I-1)(0)} = t \tag{3.12a}$$

$$\hat{r}_R^{(I-1)(0)} = P_R^t \tag{3.12b}$$

$$\hat{\dot{r}}_R^{(I-1)(0)} = \hat{\dot{h}}_R^{(I-1)(0)} \dot{P}_R^t \tag{3.12c}$$

To enable adjustment of the interacting curve, we define its candidate start and end times as $\hat{t}_S$ and $\hat{t}_F$, and constrain the interacting curve as follows:

$$\begin{aligned} \hat{h}_R^{(I)(0)} &= \hat{t}_S \\ \hat{h}_R^{(I)(n)} &= \hat{t}_F \end{aligned} \qquad t < \hat{t}_S < \hat{t}_F. \tag{3.13}$$

Fig. 3.5(a) illustrates the system state at $T = t$. This time approximately coincides with the instant at which the object state is measured, denoted by t_m, with discrepancies arising from measurement delay and sampling frequency.

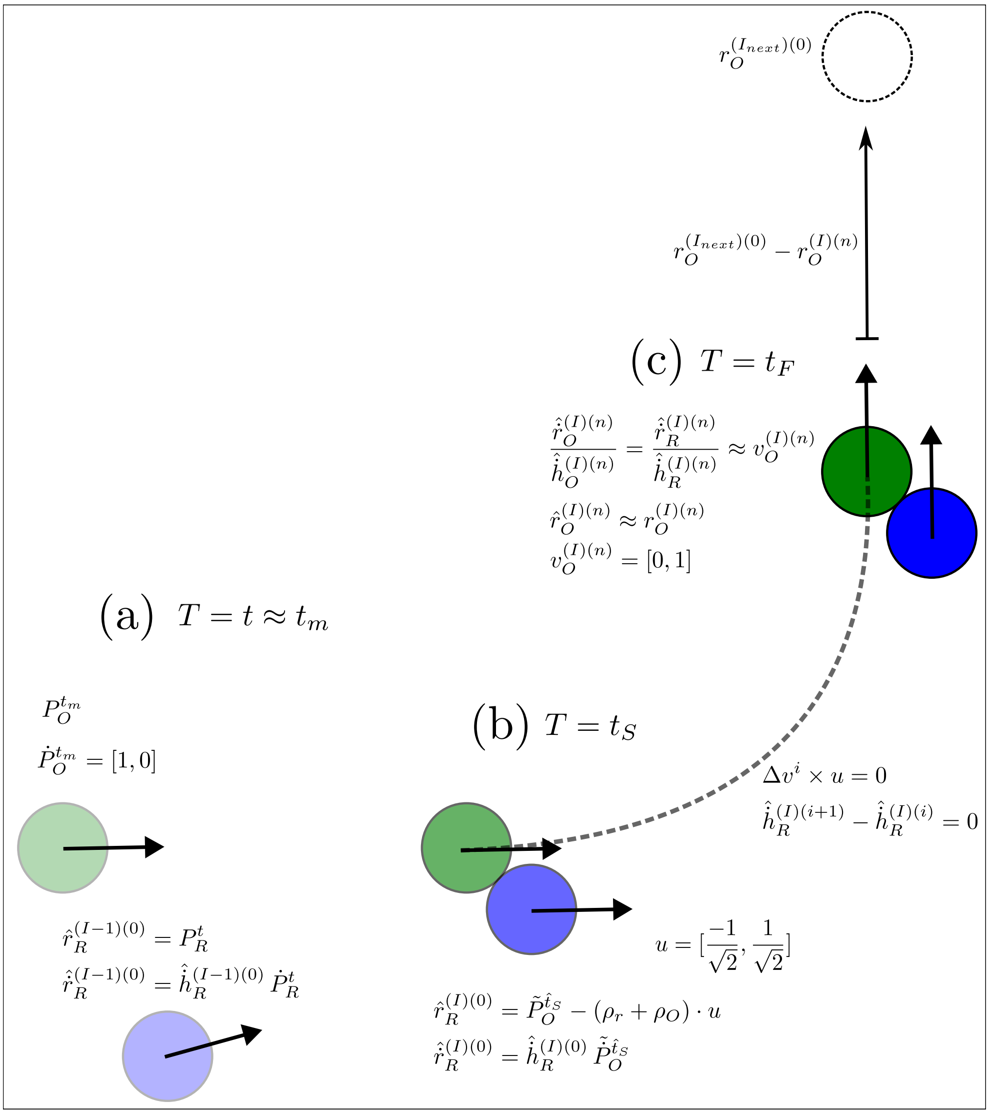

Figure 3.5: Illustrative re-planning instance in which the object's measured pre-interaction velocity is $[1, 0]$, while the offline-generated plan specifies a desired post-interaction velocity of $v_O^{(I)(n)} = [0, 1]$. (a) Time instant at which re-planning is triggered; the robot (shown in blue) and object (shown in green) states are measured nearly simultaneously, assuming negligible measurement delay. (b) Start of the interaction, where the robot has completed rendezvous with the object and is positioned at a distance $\rho_R + \rho_O$ from the object, opposite to the unit push direction u. (c) End of the interaction, at which the object has approximately reached its desired post-interaction state. The large arrow in the top-right corner indicates the planned trajectory toward the subsequent interaction location. The dashed curve denotes the arc followed by the object assuming the applied force remains parallel to u. Arrows indicate translational velocity.

3.4.3 Interaction Start

At the start of the interaction, $T = t_S$, the robot must achieve rendezvous with the object. Since re-planning is executed *prior* to an interaction, the planner must predict the object state at any candidate interaction start time $\hat{t}_S$. As the object is un-actuated and the environment is assumed to be (near) frictionless, the predicted object state, $\tilde{P}_O^{\hat{t}_S}$, is obtained using linear propagation.

$$
\begin{aligned}
\tilde{P}_O^{\hat{t}_S} &= P_O^{t_m} + \dot{P}_O^{t_m} \cdot (\hat{t}_S - t_m) && \text{(3.14a)} \\
\tilde{\dot{P}}_O^{\hat{t}_S} &= \dot{P}_O^{t_m} && \text{(3.14b)}
\end{aligned}
$$

In addition to rendezvous, the robot must be correctly positioned relative to the object to enable the transfer of positive normal forces. To formalize this requirement, the unit push direction u is defined as the normalized velocity difference between the measured object velocity and the desired post-interaction object velocity. This direction represents the ideal axis along which the application of a constant force would produce the desired change in object velocity. Note here that $v_O^{(I)(n)}$ is a known quantity representing the desired object velocity at the end of the interaction:

$$
\begin{aligned}
\Delta v &= v_O^{(I)(n)} - \tilde{\dot{P}}_O^{\hat{t}_S}, \\
u &= \frac{\Delta v}{\|\Delta v\|_2}
\end{aligned}
\tag{3.15}
$$

To ensure proper alignment at the beginning of the interaction, the following constraints are imposed. First, the robot must be positioned at a distance of $\rho_R + \rho_O$ from the predicted object position along the direction u. Second, the robot's velocity must match the predicted object velocity.

$$
\begin{aligned}
\hat{r}_R^{(I)(0)} &= \tilde{P}_O^{\hat{t}_S} - (\rho_r + \rho_O) \cdot u && \text{(3.16a)} \\
\hat{\dot{r}}_R^{(I)(0)} &= \hat{\dot{h}}_R^{(I)(0)} \quad \tilde{\dot{P}}_O^{\hat{t}_S} && \text{(3.16b)}
\end{aligned}
$$

Fig. 3.5(b) illustrates the system state at $T = t_S$ for an example scenario, highlighting the unit push direction.

3.4.4 Interaction End

In addition to ensuring proper alignment at the start of the interaction, directional consistency must be maintained throughout the interaction interval. To this end, alignment constraints are enforced along the interaction Bézier curve to ensure that the robot continues to apply force in the desired push direction.

$$\begin{aligned} &\Delta v^i = \hat{\dot{r}}_R^{(I)(i+1)} - \hat{\dot{r}}_R^{(I)(i)} \\ &\Delta v^i \times u = 0 \qquad\qquad\qquad \forall i \in [0, n-1] \\ &\hat{\dot{h}}_R^{(I)(i+1)} - \hat{\dot{h}}_R^{(I)(i)} = 0 \end{aligned} \tag{3.17}$$

Following the motivation discussed in Section 3.3.3.3, the directional constraints are applied only to the spatial derivative ($\dot{r}$), while the temporal derivative ($\dot{h}$) is held constant.

Although the object trajectories are not re-planned, the object terminal position and velocity are defined as follows:

$$\hat{r}_O^{(I)(n)} = \hat{r}_R^{(I)(n)} + (\rho_R + \rho_O) \cdot u \tag{3.18a}$$

$$\hat{\dot{r}}_O^{(I)(n)} = \hat{\dot{r}}_R^{(I)(n)} \tag{3.18b}$$

$$\hat{\dot{h}}_O^{(I)(n)} = \hat{\dot{h}}_R^{(I)(n)} \tag{3.18c}$$

Moreover, to ensure that the object, once released by the robot, reaches the subsequent interaction, the terminal direction of motion must be consistent with the direction specified by the offline plan.

Let $r_O^{(I_{next})(0)}$ denote the object position at the start of the subsequent interaction, as determined by the MICP planner, and let $r_O^{(I)(n)}$ and $v_O^{(I)(n)}$ denote the planned object position and velocity at the end of the current interaction, respectively.

Geometrically, this requirement can be interpreted as follows. Define the nominal displacement vector

$$A = r_O^{(I_{next})(0)} - r_O^{(I)(n)},$$

which specifies the direction of motion between successive interactions in the offline plan. Similarly, define the displacement vector

$$B = r_O^{(I_{next})(0)} - \hat{r}_O^{(I)(n)},$$

which represents the direction induced by the re-planned interaction. Enforcing $A \parallel B$ ensures that the re-planned terminal object position $\hat{r}_O^{(I)(n)}$ lies on the line spanned by the nominal displacement vector A, independent of its distance from the next interaction point. In addition, enforcing co-linearity between the object velocity at release and the vector A, or equivalently the planned terminal velocity $v_O^{(I)(n)}$, guarantees directional consistency of the object trajectory while still allowing deviations in timing, position, and velocity magnitude. These requirements are enforced through the following directional alignment constraints.

$$\left(r_O^{(I_{next})(0)} - r_O^{(I)(n)}\right) \times \left(r_O^{(I_{next})(0)} - \hat{r}_O^{(I)(n)}\right) = 0, \tag{3.19a}$$
$$\hat{\dot{r}}_O^{(I)(n)} \times v_O^{(I)(n)} = 0 \tag{3.19b}$$

Fig. 3.5(c) illustrates the resulting directional alignment constraints for an example interaction.

If the current interaction corresponds to the final planned interaction, i.e., $r_O^{(I_{next})(0)} = \emptyset$, and the desired terminal object velocity satisfies $v_O^{(I)(n)} = \mathbf{0}$, the constraint simplifies to

$$\hat{\dot{r}}_O^{(I)(n)} = 0 \tag{3.20}$$

Finally, to ensure that the re-planned trajectory remains as close as possible to the nominal offline plan, we introduce soft penalties on the deviations in t_S, t_F, r_O, $\dot{r}_O$ and $\dot{h}_O$. Incorporating these terms, we formulate the following Quadratic Program (QP):

$$\begin{aligned}
\delta_{t_1} &= \|\hat{t}_S - t_S\|_2^2, \\
\delta_{t_2} &= \|\hat{t}_F - t_F\|_2^2, \\
\delta_r &= \|\hat{r}_O^{(I)(n)} - r_O^{(I)(n)}\|_2^2, \\
\delta_{\dot{r}} &= \|\hat{\dot{r}}_O^{(I)(n)} - \dot{r}_O^{(I)(n)}\|_2^2, \\
\delta_{\dot{h}} &= \|\hat{\dot{h}}_O^{(I)(n)} - \dot{h}_O^{(I)(n)}\|_2^2,
\end{aligned}$$

$$
\begin{aligned}
\min_{r_R, h_R} \quad & \ddot{r}_R(s)^\top Q_{\ddot{r}} \ddot{r}_R(s) + \ddot{h}_R(s)^\top Q_{\ddot{h}} \ddot{h}_R(s) + \delta_{t_1} + \delta_{t_2} + \delta_r + \delta_{\dot{r}} + \delta_{\dot{h}}, \\
\text{s.t.} \quad & \\
& \text{Eqs. (3.1c), (3.3), (3.8), (3.12), (3.13), (3.16)–(3.18).} \\
& \begin{cases} \text{Eq. (3.19)}, & v_O^{(I)(n)} \neq \mathbf{0} \ \wedge \ r_O^{(I_{next})(0)} \neq \emptyset, \\ \text{Eq. (3.20)}, & v_O^{(I)(n)} = \mathbf{0} \ \wedge \ r_O^{(I_{next})(0)} = \emptyset. \end{cases}
\end{aligned}
\tag{3.21}
$$

3.5 MPC

3.5.1 Approach

The purpose of the MPC is to enable real-time control of the robots, allowing the transportation problem to be executed on a physical free-flyer platform. Since the overall task can be divided into two distinct phases, interaction and non-interaction, we introduce two controllers: a nominal controller and a coupled controller. The nominal MPC governs trajectories without interactions and ensures accurate tracking of the planned motion. In contrast, the coupled MPC explicitly models robot–object interactions by coupling their dynamics, this provides low-level feedback to ensure that the object follows its reference trajectory during contact.

3.5.2 Force and Rate Model of a Free-Flyer

In this work, the robots are represented as rigid bodies operating in the 2D plane and actuated through forces and rate inputs, consistent with the formulation in [14]. Let m denote the robot mass. The free-flyer dynamics are given by

$$\dot{p} = v, \tag{3.22a}$$

$$\dot{v} = \frac{1}{m} R(q)^T F, \tag{3.22b}$$

$$\dot{q} = \frac{1}{2} \Xi(q)\, \tau, \tag{3.22c}$$

where the robot state is $x_R = [p, v, q]$ and the input vector is $u = [F^T, \tau]$. Here, $F \in \mathbb{R}^2$ is the applied force in the body frame, and $\tau \in \mathbb{R}$ is the body yaw rate input. The matrix $R(q)$ denotes the rotation from the body frame to the world frame, while $\Xi(q)$ maps angular velocity to the corresponding quaternion rate.

This modeling approach captures translational motion and orientation, but omits explicit rotational velocity states. Instead, the orientation evolves directly according to a rate-driven quaternion kinematic equation

3.5.3 Nominal MPC

To ensure that the robots accurately track their reference trajectories (Bézier curves), we define the nominal MPC problem as follows:

$$
\begin{aligned}
J^*(\tilde{x}(k)) = \min_{\mathbf{u}_k} & \; J(x(n|k), u(n|k)), && \text{(3.23a)}\\
\text{s.t.} & \\
x(m+1|k) &= g(x(m|k), u(m|k)), && \text{(3.23b)}\\
x(m|k) &\in \mathbb{X}, && \text{(3.23c)}\\
u(m|k) &\in \mathbb{U}, \quad \forall m \in \mathbb{N}_{[0,N-1]}, && \text{(3.23d)}\\
x(0|k) &= x(k), \quad \forall n \in \mathbb{N}_{[0,N]}, && \text{(3.23e)}
\end{aligned}
$$

where $g(x(m|k), u(m|k)) = \text{RK4}[(3.22a) - (3.22c)]$ represents the discrete-time dynamics of the system. Since neither planner explicitly handles collision avoidance between robots and objects outside of interactions, following the method in [14], safety during non-interacting trajectories is enforced using a Control Barrier Function (CBF) [34].

The cost function (3.23a) is described in detail in Appendix B, but in essence, for each shooting stage $m \in \mathbb{N}_{[0,N]}$, we sample the robot's Bézier curve to construct a reference trajectory

$$\mathbf{x}_R^{\text{ref}} = [\mathbf{p}_R^{\text{ref}}, \mathbf{v}_R^{\text{ref}}, \mathbf{q}_R^{\text{ref}}] \tag{3.24}$$

which is then used in the cost function. Since neither the MICP planner nor the online planner considers orientation explicitly, we define q^{ref} as the zero quaternion at all stages.

3.5.4 Coupled MPC

To describe the interaction between the robot and the object (see Fig 3.6), we model the robot as directly influencing the object's linear accelerations.

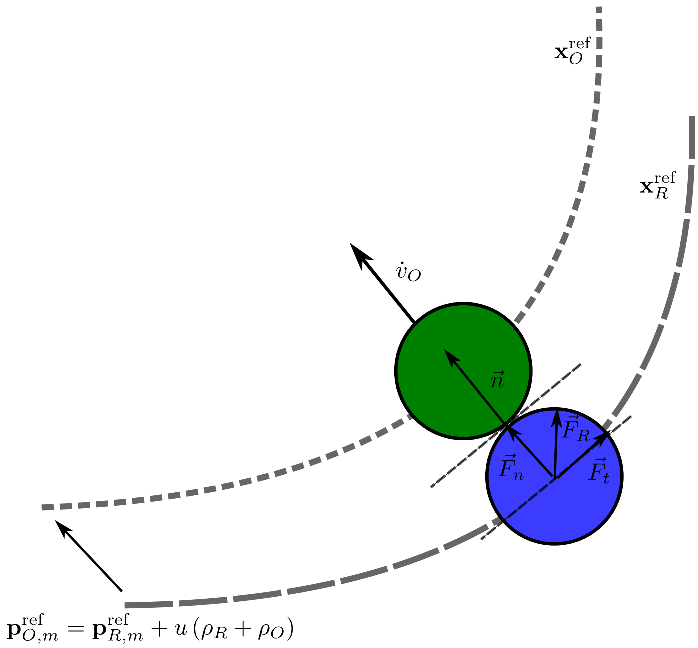

Figure 3.6: Example interaction between a robot (shown in blue) and an object (shown in green) using the coupled MPC, illustrating the synthesis of the object reference trajectory and the resulting force transfer during interaction.

Angular acceleration and orientation of the object are not included, as their incorporation would considerably complicate the formulation. To facilitate the control of the object, we introduce a reference trajectory $\mathbf{x}_O^{\text{ref}}$ to be used in the cost function of the coupled controller. The reference trajectory is constructed as follows, where we reiterate that $v_O^{(I)(n)}$ denotes the desired velocity of the object at the end of the interaction.

$$\begin{aligned}
\Delta v &= v_O^{(I)(n)} - \dot{P}_O \\
u &= \frac{\Delta v}{\|\Delta v\|_2} \\
\mathbf{p}_{O,m}^{\text{ref}} &= \mathbf{p}_{R,m}^{\text{ref}} + u\,(\rho_R + \rho_O) \\
\mathbf{v}_{O,m}^{\text{ref}} &= \mathbf{v}_{R,m}^{\text{ref}}
\end{aligned} \qquad \forall m \in \mathbb{N}_{[0,N]} \tag{3.25}$$

Here, u denotes the unit push direction. The object velocity is set equal to the robot velocity, and the object reference trajectory is obtained by offsetting the robot reference trajectory by $\rho_R + \rho_O$ along u (see Fig. 3.6). The state vector of the coupled MPC consists of the concatenated robot state $x_R = [p_R, v_R, q_R]$ and object state $x_O = [p_O, v_O]$. Using the constructed trajectory in (3.25), we define the sampled reference trajectory of the object as

$$\mathbf{x}_O^{\text{ref}} = [\mathbf{p}_O^{\text{ref}}, \mathbf{v}_O^{\text{ref}}] \tag{3.26}$$

This reference includes only position and velocity, since the object's orientation is not modelled.

The interaction dynamics are based on the state derivatives in (3.22a)–-(3.22c). Equation (3.22a) remains unchanged as it lacks inputs, and (3.22c) is unchanged due to the omission of object orientation. Substituting $m = 1$ into (3.22b) yields the robot-generated force in the world frame, denoted by $\vec{F}_R$. The resulting interaction model is

$$\vec{\xi} = \vec{p}_O - \vec{p}_R \tag{3.27a}$$

$$\delta = \nu(\|\vec{\xi}\|_2 - \rho_R - \rho_O - \gamma) \tag{3.27b}$$

$$\eta = \frac{1 - \tanh(\delta/2)}{2} \tag{3.27c}$$

$$\vec{n} = \frac{\vec{\xi}}{\|\vec{\xi}\|_2} \tag{3.27d}$$

$$p_s = \eta\,(\vec{n} \cdot \vec{F}_R) \tag{3.27e}$$

$$P_s = \frac{1}{2}(p_s + \sqrt{p_s^2}) \tag{3.27f}$$

$$\vec{F}_n = \vec{n}\,P_s \tag{3.27g}$$

$$\dot{v}_O = \vec{F}_n \,/\, m_O \tag{3.27h}$$

$$\dot{v}_R = \frac{\vec{F}_R - \vec{F}_n}{m_R} \tag{3.27i}$$

Here, $\nu = 100$ is the activation gain, $\gamma = 0.005$ is the proximity threshold, ρ_R and ρ_O are the robot and object radii, and m_R, m_O their masses. Equations (3.27a)–-(3.27c) define the activation function η, which signals when the robot is sufficiently close to transfer force (Fig. 3.7). Equation (3.27e) projects $\vec{F}_R$ onto the normal $\vec{n}$ and is modulated by η, while (3.27f) ensures only positive normal forces are applied. Equations (3.27g)–-(3.27i) describe the transmitted force and the resulting accelerations of the object and robot. These interaction dynamics are then incorporated into the system model $g(x(m|k), u(m|k))$, enabling the formulation of the coupled MPC problem as given in (3.23a)–-(3.23e).

The complete cost function of the coupled controller is described in detail in Appendix B.

Example 4. *Consider the example interaction shown in Fig. 3.6. Using the robot reference trajectory* $\mathbf{x}_R^{ref}$ *and* (3.25)*, the object reference trajectory* $\mathbf{x}_O^{ref}$ *is constructed and used in the cost function of the coupled MPC (see Appendix B). When the centre-to-centre distance between the robot and the object is sufficiently small, the activation function* η *is active, as illustrated in Fig. 3.7. As a result, the projected normal force* $\vec{F}_n$*, defined in* (3.27e)–(3.27g)*, is non-zero, allowing the controller to model the induced object acceleration according to* (3.27h).

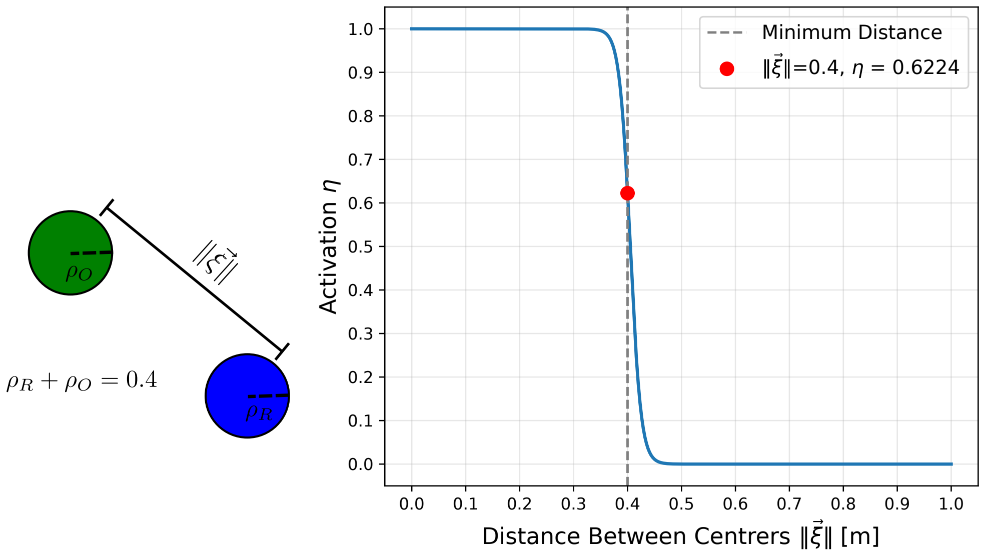

Figure 3.7: Example illustrating the relationship between the activation function η and the centre-to-centre distance $|\vec{\xi}|$, as defined in (3.27b)–(3.27c), with $\rho_R = \rho_O = 0.2$.

Chapter 4

Results and Analysis

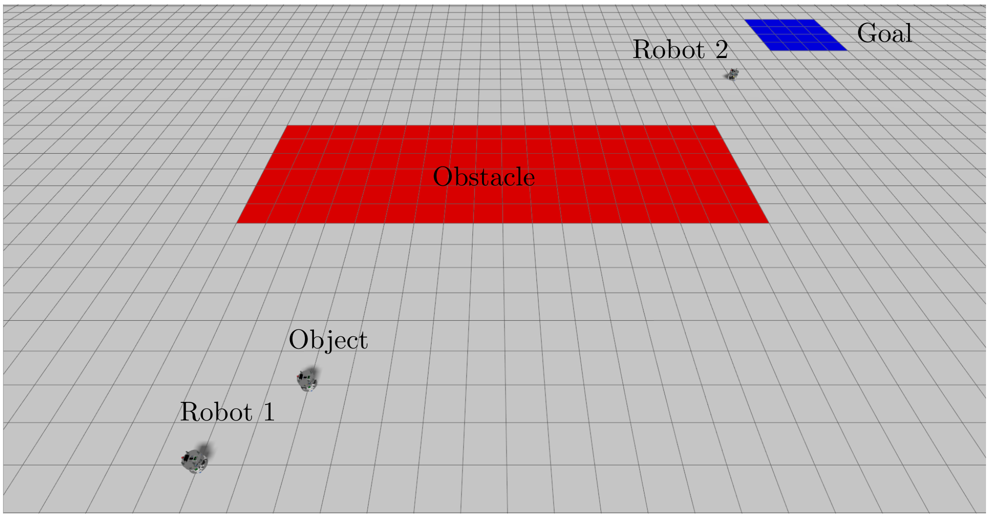

Figure 4.1: High-fidelity Gazebo simulation environment representing the ATMOS platform used for all simulation experiments.

To evaluate the proposed method, experiments are conducted in a simulated two-dimensional microgravity analogue environment based on the Autonomy Testbed for Multi-purpose Orbiting Systems (ATMOS) [35] platform at KTH Royal Institute of Technology (KTH). This platform consists of robots supported by air bearings to emulate the microgravity conditions of space, and actuated by pneumatic thrusters that enable both linear and angular motion in two dimensions.

All simulations are performed using a high-fidelity simulator that accurately captures the physical behaviour of the platform (Fig. 4.1). One notable simplification is the assumption of frictionless interaction between the robots. This assumption is made primarily to simplify the modelling process, as accurately modelling the physics of the air bearings would significantly increase the complexity of the simulation.

In addition to the simulation studies, the results are complemented by a hardware evaluation of the proposed method conducted on the ATMOS platform.

Position and velocity measurements are obtained from either ground-truth data in Gazebo or the motion capture system on the ATMOS platform. We utilize Gurobi [36] to solve the MIP, CasADi [37] for re-planning, and acados [38] for the MPC. All computations are performed centrally on a laptop with an Intel Core i7-1360P CPU and 16GB of RAM.

4.1 Simulation Results

A total of four transportation scenarios with varying levels of difficulty are simulated. For all scenarios, the nominal MPC is employed during non-interacting curves. To better highlight the effect of the coupled controller, each scenario is executed for a total of 20 iterations. 10 iterations employ the coupled MPC for interacting curves, while the remaining 10 iterations use the nominal controller for both interacting and non-interacting curves.

For evaluation purposes, six performance metrics are considered, as summarized in Table 4.1. These metrics are selected to capture complementary aspects of task execution and to assess the effectiveness of the proposed method at different stages of the interaction. Rendezvous accuracy is quantified by Δv_{pre}, which captures the velocity mismatch between the robot and the object at the time of contact. The accuracy of the post-interaction outcome is evaluated using Δv_{post}, measuring deviation from the desired object velocity at the end of the interaction. Task-level performance is assessed through the final object velocity and position errors, Δv_{Final} and Δp_{Final}, which indicate how accurately the overall task objectives are achieved. Finally, the metrics Δ_T and $\Delta\rho$ quantify deviations from the nominal task duration and robustness margin, respectively. For Scenario 1, the metric ρ is omitted, as no obstacles are present. All metrics, with the exception of Δv_{pre}, are evaluated relative to the offline-generated MICP plan.

Δv_{pre}	Velocity difference between robot and object at rendezvous
Δv_{post}	Object post-interaction velocity deviation
Δv_{Final}	Object velocity deviation at final time T
Δp_{Final}	Object position deviation at final time T
ΔT	Total task time deviation
$\Delta \rho$	Spatial robustness deviation

Table 4.1: Metrics

4.1.1 Basic Transportation

Shown in Fig. 4.2(a) is the setup for Scenario 1. The transportation tasks involves robots R_1 and R_2 transporting the object O_1 from its starting position to the goal. This scenario excludes any obstacles or RoI, as such no STL formulas are used. The total allotted time for the task is $T = 65$ seconds and the initial conditions are as follows: $R_1(0) = [2, 2]$, $R_2(0) = [20, 20]$, $O_1(0) = [4, 4]$. The goal of the task is $O_1(T) = [25, 25]$. An illustrative example of the original, re-planned and realized trajectories using the coupled controller is shown in Figures 4.2(b)–4.2(d).

The performance metrics for this scenario are presented in Table 4.2. The low values of Δv_{pre} observed for both controllers indicate gentle contact between the robot and the object during the rendezvous phase. A comparison between the data collected using the coupled and nominal MPC, respectively, shows that the coupled MPC more effectively achieves the desired object post-interaction velocity. Furthermore, the coupled controller demonstrates improved accuracy in guiding the object to its desired goal position, indicating superior interaction handling and final positioning performance in this scenario.

	Coupled MPC		**Nominal** MPC	
	Mean	Median	Mean	Median
Δv_{pre}	0.014 m/s	0.013 m/s	0.054 m/s	0.023 m/s
Δv_{post}	0.076 m/s	0.073 m/s	0.213 m/s	0.128 m/s
Δv_{Final}	0.078 m/s	0.066 m/s	0.307 m/s	0.227 m/s
Δp_{Final}	0.469 m	0.427 m	3.140 m	3.128 m
ΔT	10.230 s	10.336 s	11.300 s	12.135 s
$\Delta \rho$	-	-	-	-

Table 4.2: Metrics Scenario 1

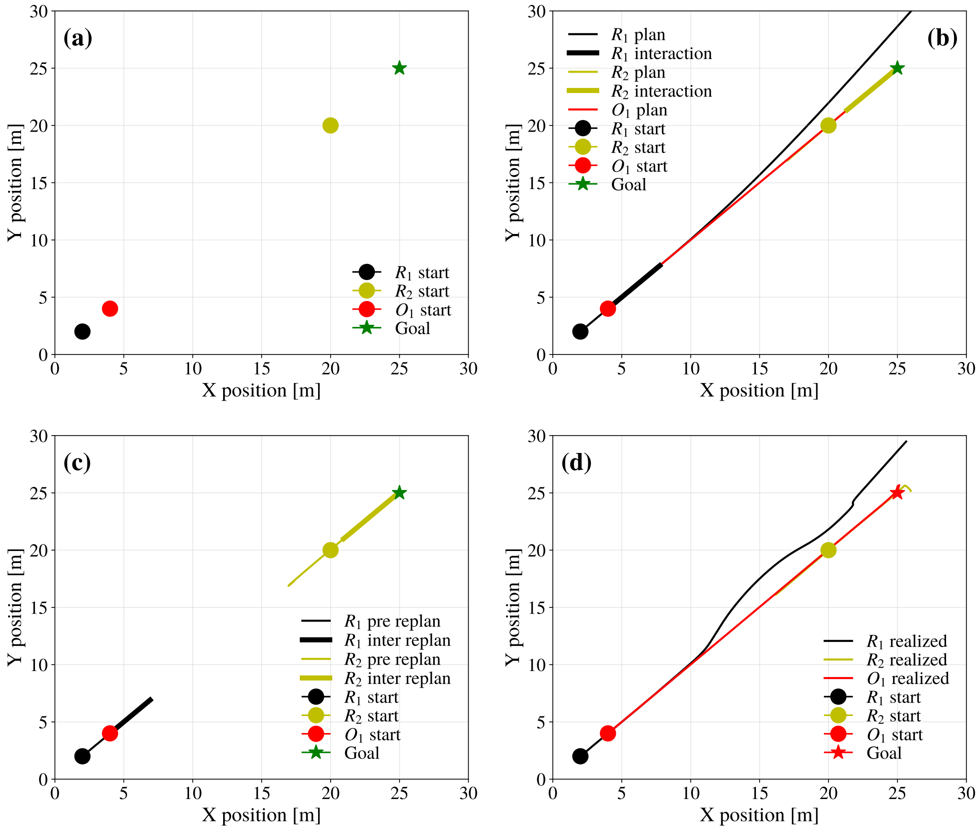

Figure 4.2: Example realization of Scenario 1 showing (a) the task setup, (b) the nominal plans generated by the MICP, (c) the re-planned robot trajectories during task execution, and (d) the final realized trajectories of all robots and the object.

4.1.2 Transportation & Obstacle Avoidance

To evaluate the performance of the proposed method on non-linear trajectories, two scenarios are designed and simulated, each containing an obstacle located between the initial and goal positions of the object. These scenarios are intended to assess both the method's ability to operate in environments with obstacles and hazards, as well as its performance when interactions induce changes in the object's direction rather than simple acceleration or deceleration along a straight-line path.

Both scenarios share similar initial conditions for the robots and the object, given by $R_1(0) = [2, 2]$, $R_2(0) = [20, 20]$, and $O_1(0) = [4, 4]$ for Scenario 2 and $R_1(0) = [1, 4]$, $R_2(0) = [2, 9]$, and $O_1(0) = [3, 3]$ for Scenario 3.

The corresponding setups are shown in Figures 4.3(a) and 4.4(a), with task horizons of $T = 100$ and $T = 90$ for Scenarios 2 and 3, respectively. The objective in both tasks is for the object to reach the final goal position $O_1(T) = [25, 25]$, while ensuring obstacle avoidance for all systems throughout the execution:

$$\varphi = \bigwedge_{S_i \in \mathcal{S}} \mathbf{G}_{[0,T]}(p_{S_i} \in \mathcal{W}_{free})$$

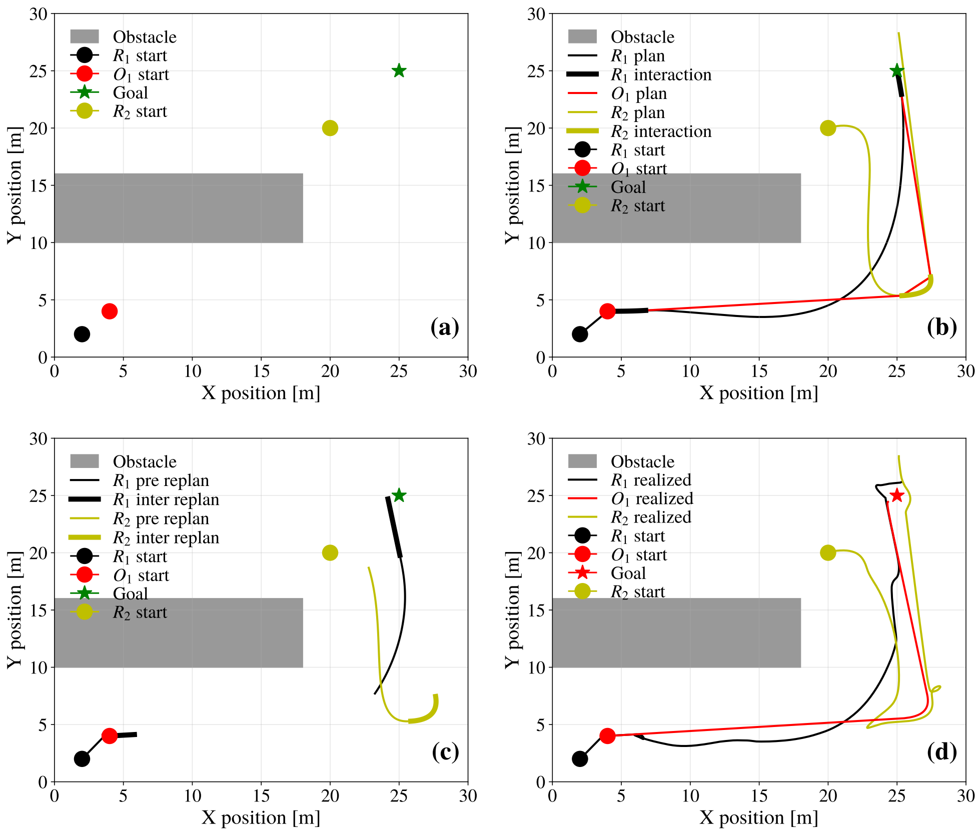

Figure 4.3: Example realization of Scenario 2 showing (a) the task setup, (b) the nominal plans generated by the MICP, (c) the re-planned robot trajectories during task execution, and (d) the final realized trajectories of all robots and the object.

Figures 4.3(b)–-4.3(c) and 4.4(b)–-4.4(c) present representative comparisons between the nominal plans and the re-planned trajectories for scenarios 2 and 3, respectively. In both cases, the results illustrate the effect of the online planner accounting for the physical size of the objects and robots as well as their updated position and velocity. This leads to a shift in the interaction locations while still aiming to minimize deviations of the object's trajectory from the originally planned path.

Similar to Scenario 1, the metrics reported in Tables 4.3 and 4.4 indicate improved performance when using the coupled controller. In both scenarios, the coupled MPC achieves smaller deviations in Δv_{post} and Δp_{Final} relative to the original plans compared to the nominal controller.

The comparison of $\Delta\rho$, however, is less straightforward, as neither the coupled nor the nominal MPC exhibits a clear advantage. This behaviour is expected since neither the re-planner nor the MPC explicitly addresses obstacle avoidance. Instead, $\Delta\rho$ primarily serves as an indicator of the overall deviation of the executed trajectories from the original plans. While a small number of outliers lead to larger deviations in some cases, the results suggest that the difference in robustness between the planned and realized trajectories remains minor overall. A closer inspection of Figures 4.3(d) and 4.4(d) reveals several large perturbations in the realized robot trajectories when compared to the original plans. Notably, Fig. 4.4(d) shows a pronounced kink in the trajectory of R_1. This behaviour arises from the Control Barrier Function, which acts as a safety filter on the control inputs and enforces a minimum separation distance of 1 m between the robot and the object outside interaction phases.

	Coupled MPC		**Nominal** MPC	
	Mean	Median	Mean	Median
Δv_{pre}	0.019 m/s	0.017 m/s	0.053 m/s	0.023 m/s
Δv_{post}	0.105 m/s	0.063 m/s	0.152 m/s	0.202 m/s
Δv_{Final}	0.040 m/s	0.037 m/s	0.050 m/s	0.009 m/s
Δp_{Final}	0.814 m	0.822 m	6.116 m	6.297 m
ΔT	25.479 s	24.951 s	26.242 s	23.369 s
$\Delta\rho$	-0.001	-0.001	−0.120	−0.003

Table 4.3: Metrics Scenario 2

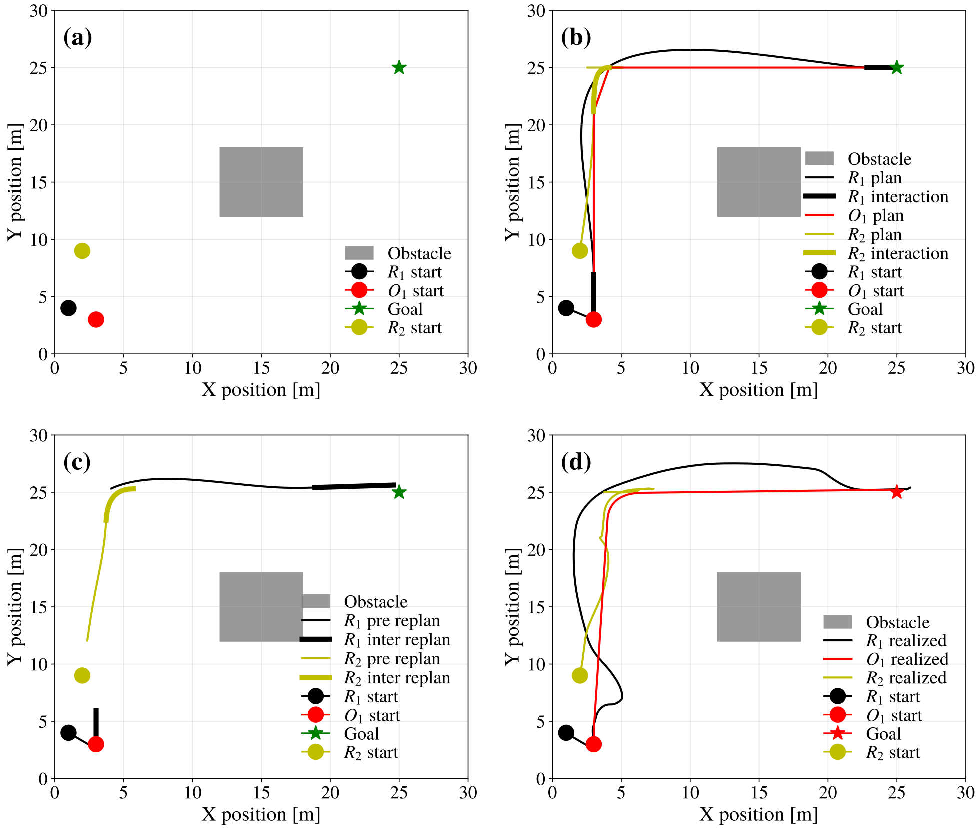

Figure 4.4: Example realization of Scenario 3 showing (a) the task setup, (b) the nominal plans generated by the MICP, (c) the re-planned robot trajectories during task execution, and (d) the final realized trajectories of all robots and the object.

	Coupled MPC		**Nominal** MPC	
	Mean	Median	Mean	Median
Δv_{pre}	0.030 m/s	0.022 m/s	0.061 m/s	0.038 m/s
Δv_{post}	0.260 m/s	0.073 m/s	0.329 m/s	0.211 m/s
Δv_{Final}	0.058 m/s	0.056 m/s	0.024 m/s	0.016 m/s
Δp_{Final}	0.670 m	0.515 m	4.412 m	3.332 m
ΔT	32.173 s	32.925 s	44.996 s	46.784 s
$\Delta \rho$	-0.523	+0.009	–0.108	–0.094

Table 4.4: Metrics Scenario 3

4.1.3 Transportation with Waypoint Regions

The final scenario evaluates the performance of the proposed method in the presence of intermediary goals during transport, which also serves to assess its behaviour under a more complex STL specification. Fig. 4.5 illustrates the setup for Scenario 4. The initial conditions are $R_1(0) = [2, 2]$, $R_2(0) = [1, 5]$, and $O_1(0) = [2, 4]$. The task is to transport O_1 to the goal position at $T = 122$, $O_1(T) = [25, 25]$, while visiting two regions of interest, $RoI_1 = [10, 7]$ and $RoI_2 = [19, 21]$, each defined as a square of side length 2. The specification is given by

$$\varphi = \mathbf{F}_{[0,60]}(p_{O_1} \in RoI_1) \wedge \mathbf{F}_{[80,T]}(p_{O_1} \in RoI_2)$$

The metrics shown in Table 4.5 indicate comparable performance between the coupled and nominal controllers, with a slight robustness advantage observed for the nominal controller. In contrast to the other scenarios, no metric demonstrates a consistent advantage of the coupled MPC over the nominal formulation.

	Coupled MPC		**Nominal** MPC	
	Mean	Median	Mean	Median
Δv_{pre}	0.023 m/s	0.016 m/s	0.031 m/s	0.020 m/s
Δv_{post}	0.026 m/s	0.026 m/s	0.047 m/s	0.027 m/s
Δv_{Final}	0.037 m/s	0.043 m/s	0.082 m/s	0.012 m/s
Δp_{Final}	0.552 m	0.560 m	1.108 m	0.500 m
ΔT	5.481 s	5.170 s	4.693 s	3.711 s
$\Delta \rho$	-0.107	-0.110	–0.082	–0.037

Table 4.5: Metrics Scenario 4

4.2 Simulation Analysis

4.2.1 High-Level Performance Trends

An overview of all performance metrics across the evaluated scenarios reveals a consistent advantage of the coupled MPC. In particular, the coupled controller achieves lower mean values and reduced variance for the interaction-related metric Δv_{post}, and the task-level metric Δp_{Final}. These improvements are most pronounced in scenarios that require higher object accelerations

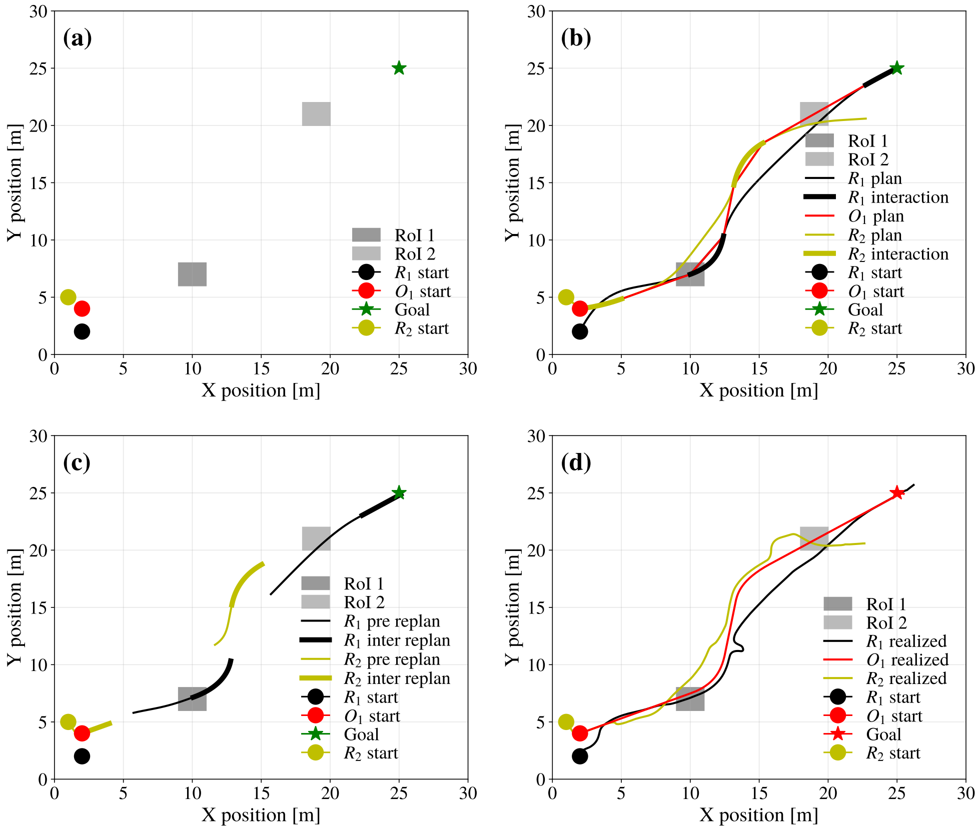

Figure 4.5: Example realization of Scenario 4 showing (a) the task setup, (b) the nominal plans generated by the MICP, (c) the re-planned robot trajectories during task execution, and (d) the final realized trajectories of all robots and the object.

(Scenario 1) or induce directional changes in the object motion (Scenarios 2 and 3). In contrast, Scenario 4, characterized by low interaction forces and gradual object motion, exhibit smaller performance differences between the two controllers. Beyond lower mean errors, the reduced variance observed with the coupled MPC indicates more consistent interaction outcomes across executions.

4.2.2 Interaction Quality and Momentum Transfer

Rendezvous

A key indicator of interaction quality is the velocity difference between the robot and the object at rendezvous, quantified by Δv_{pre}. For this metric, both controllers consistently achieve low mean and variance across all scenarios. This is to be expected as rendezvous is modelled identically in both controllers, highlighting that the framework is capable of ensuring a gentle and precise rendezvous before interaction phases.

Post-interaction Velocity

The advantage of the coupled MPC becomes even more pronounced when examining the post-interaction velocity deviation Δv_{post}. Across all scenarios, the coupled controller exhibits a considerable reduction in both the mean and variance of this metrics, as compared to the nominal formulation. This suggests that the coupled controller more accurately anticipates the momentum transfer and is better able to control the object's trajectory during an interaction, and as a result, its post-interaction trajectory. This is particularly evident in Scenarios 2 and 3, which require directional changes of the object, as opposed to just straight-line motion. This advantage is largely attributed to the controllers ability to model the interaction and the required forces, allowing it to apply control inputs that minimize the deviation of both itself and the object from their respective trajectories. In contrast, the nominal controller does not consider the object or the dynamics required for interactions. Its control input only aims to ensure accurate tracking of its own trajectory. Consequently, there is a higher chance of separation, slipping or harsh control inputs during interaction phases, resulting in an overall larger mean and variance of Δv_{post} across most scenarios.

4.2.3 Final Object State Accuracy

Position Accuracy

The final position deviation Δp_{Final} represents the cumulative positional error of the object as compared to the original plan. This metric is perhaps the most critical task-level metric, as a successful mission ultimately depends on delivering the object to its desired goal location. In Scenarios 1–3, the coupled MPC achieves an order-of-magnitude improvement in this metric compared to the nominal controller, with the largest gains observed in scenarios involving directional changes.

These results indicate that errors introduced during interaction phases dominate the final positional error of the object. Once such errors occur, neither the online re-planner nor the MPC can fully compensate for them retroactively. For instance, if an interaction imparts an incorrect post-interaction velocity shortly before task completion, the controller can at best decelerate the object along its resulting trajectory, but cannot redirect it sufficiently to ensure convergence to the desired goal within a tight tolerance. Consequently, limiting errors during interaction phases is essential for achieving accurate final positioning. The coupled controller's improved performance during interactions and subsequent low Δv_{post} deviation, therefore, plays a critical role in reducing error accumulation over successive interactions.

Velocity Accuracy at Task Completion

The final velocity deviation Δv_{Final} exhibits a less uniform trend across scenarios. In several cases, particularly Scenarios 2–4, the nominal MPC achieves comparable or lower final velocity errors than the coupled formulation. This behaviour can be explained by the structure of the tasks, in which the desired final velocity of the object is zero. As a result, the final interaction always involves decelerating the object to rest.

While the coupled controller provides increased flexibility during interaction phases, this flexibility is most beneficial when the object initially has low momentum or when directional changes are required. In scenarios dominated by deceleration, the nominal controller can perform competitively, provided that the imparted object velocity prior to the final interaction is not excessively large. This effect is clearly illustrated in Scenario 1, where the nominal MPC imparts a significantly higher object velocity due to unmodeled dynamics, resulting in a larger deceleration demand during the final interaction and consequently a larger Δv_{Final}. In contrast, the coupled controller limits the imparted velocity earlier in the task, reducing the burden on the final interaction.

4.2.4 Task Duration

Across all scenarios and both controllers, a significant deviation in task duration ΔT is observed, in some cases exceeding 50% of the nominal task horizon. This behaviour is primarily attributed to the absence of explicit acceleration constraints in both the offline planning and online control stages. As discussed in Section 3.3.3.5, direct modeling and constraint enforcement on accelerations is not feasible within the adopted framework. Instead, accelerations are indirectly attenuated through cost terms.

To ensure smooth and stable interactions, accelerations are further penalized in the online re-planner, resulting in lower object velocities and consequently longer execution times. This accounts for a substantial portion of the increased ΔT observed for the coupled controller, as well as part of the reduction in Δv_{post}. A similar effect is present for the nominal controller; however, its inability to model object dynamics leads to larger post-interaction velocity deviations, which in turn further prolong task execution. While the resulting increase in task duration represents a limitation of the approach, it reflects a deliberate trade-off between temporal efficiency and the robustness and feasibility of physical interactions.

4.2.5 Spatial Robustness

As discussed earlier, the spatial robustness metric $\Delta\rho$ does not provide a direct measure of the STL satisfaction performance of the online planner or the MPC. Since neither explicitly enforces STL constraints, $\Delta\rho$ instead captures the deviation of the realized trajectories from the robustness margins of the original plans. The results indicate that our method manages to preserve STL satisfaction margins reasonably well across multiple scenarios. The observed outliers, indicated by the order-of-magnitude difference between mean and median, are likely caused by the activation of the CBF, which can cause robots to deviate significantly from their trajectories in order to maintain safety constraints.

4.2.6 Regime-Dependent Effectiveness

In contrast to Scenarios 1–3, both controllers exhibit similar performance across most metrics for Scenario 4. Although no definitive conclusion can be drawn, a plausible explanation is that tasks characterized by low interaction forces and small object accelerations can be handled effectively by the nominal MPC. When interactions require lower forces, the mismatch between the required and modelled interaction forces remains small. Consequently, the feedback in the controller causes gentler adjustments and control inputs that have a lower chance of causing separation or slipping.

These results suggest that the benefits of the coupled controller scale with interaction intensity and object acceleration. For tasks involving minimal acceleration and gentle interactions, the nominal controller may be sufficient or even preferable due to its lower complexity, whereas the coupled formulation provides clear advantages in more dynamically demanding regimes.

4.3 Hardware Experiments

To evaluate the proposed method beyond simulation, experiments were conducted on the ATMOS platform at KTH, as shown in Fig. 4.6. Due to the limited workspace of the platform, only a single transportation scenario was executed, as this was the only configuration realizable without modifying the physical setup. Moreover, hardware and platform limitations restrict the repeatability across experimental runs. As a result, this section does not yet report quantitative performance metrics and instead focuses on qualitative evaluation through a proof-of-concept study.

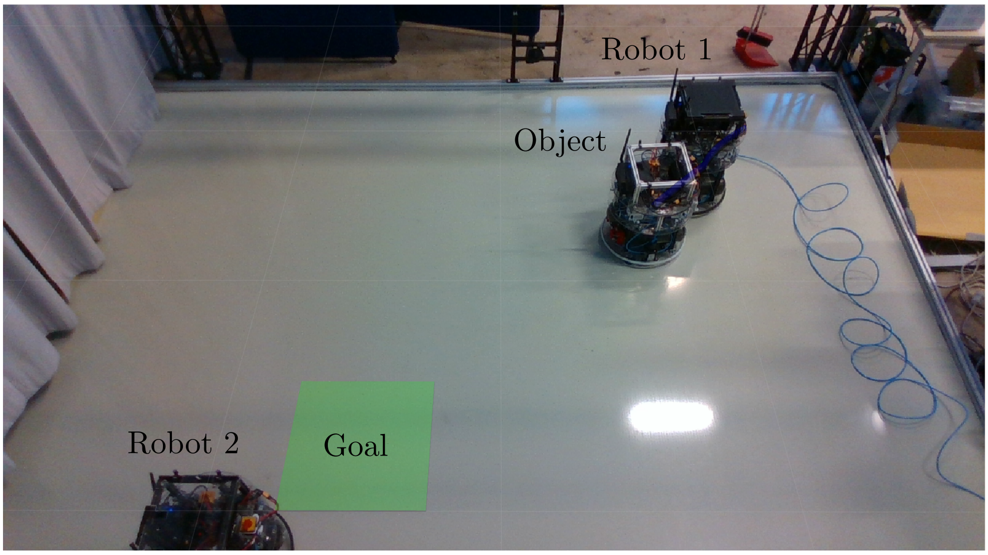

Figure 4.6: ATMOS platform at KTH used for hardware experiments.

The experiment consists of a basic transportation task equivalent to Scenario 1, scaled to fit the constraints of the ATMOS platform. Two robots, R_1 and R_2, are tasked with transporting a single object O_1 from its initial position to the target location $O_1(T) = [2.5, 0.75]$. The experimental setup is illustrated in Fig. 4.7(a), with initial conditions $R_1(0) = [0.5, -1.25]$,

$R_2(0) = [3.25, 1.5]$, $O_1(0) = [1.25, -0.5]$, and a total task duration of $T = 32$.

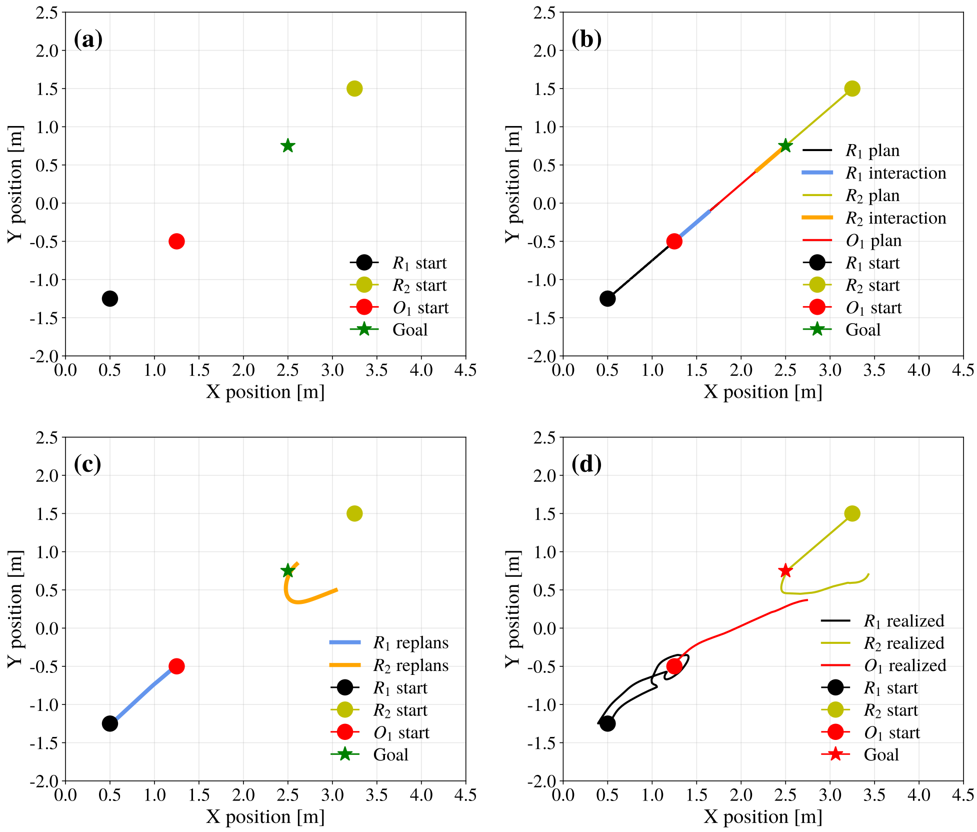

Figure 4.7: Results from hardware run 1 showing (a) the experimental setup, (b) the nominal plans generated by the MICP, (c) the re-planned robot trajectories during task execution, and (d) the final realized trajectories of all robots and the object.

The nominal plans generated by the MICP are shown in Fig. 4.7(b). Figures 4.7(c)–(d) and Figures 4.8–4.9 show the corresponding re-planned paths and realized trajectories obtained during execution. While the limited repeatability of the experiments prevents a meaningful quantitative evaluation, all experimental runs successfully transported the object to within 0.5 m of the goal region. These results qualitatively confirm that the proposed planning and re-planning framework can be executed on real hardware and remains robust to real-world disturbances, sensing noise, and execution imperfections.

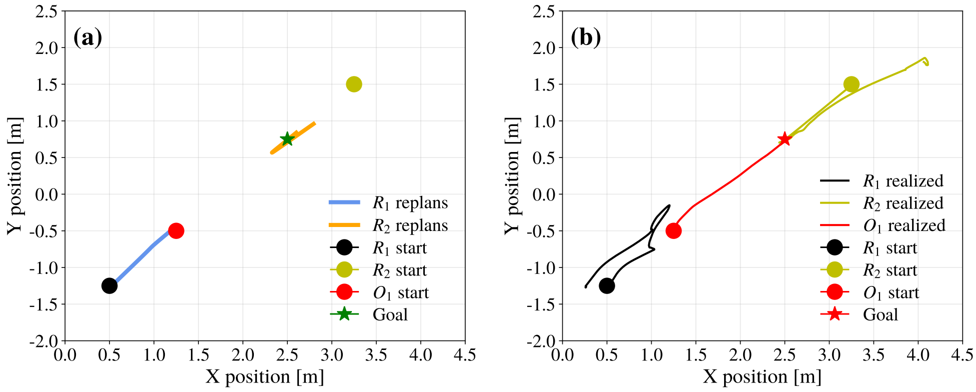

Figure 4.8: Results from hardware run 2 showing (a) the re-planned robot trajectories during task execution and (b) the final realized trajectories of all robots and the object.

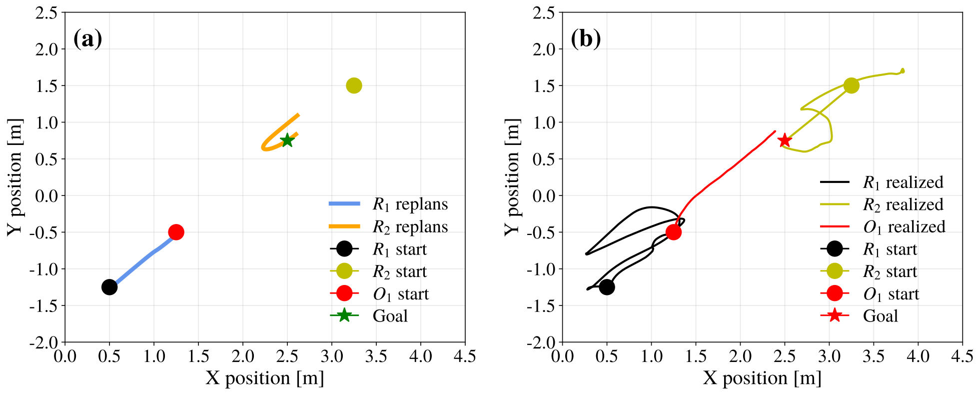

Figure 4.9: Results from hardware run 3 showing (a) the re-planned robot trajectories during task execution and (b) the final realized trajectories of all robots and the object.

4.4 Experiment Analysis

The hardware experiments execute the complete pipeline end-to-end on hardware, including sensing, offline planning, online re-planning, and trajectory execution. To reduce state uncertainty and isolate the behaviour of the planning and re-planning framework, object and robot state measurements are obtained using an external motion capture system rather than onboard state estimation. No components of the framework were otherwise adjusted or simplified for the experiments, ensuring that the observed behaviour reflects realistic system integration as close as possible.

Compared to the simulation results, the realized trajectories exhibit increased inconsistency and reduced smoothness, primarily due to actuation delays, unmodeled contact dynamics, and measurement delays from the motion capture system. Furthermore, as a result of the restricted workspace, re-planning is triggered under less ideal conditions than in simulation. This caused noticeable deviations from the nominal plans. Despite these effects, successful task completion was observed in multiple experimental runs, demonstrating that the proposed framework can operate on real hardware under non-ideal conditions.

As a final note, we would like to highlight that the experimental setup was not tuned for optimal performance on the ATMOS platform, and no platform-specific parameter optimization was performed. The goal of this experiment is therefore not to demonstrate peak efficiency or accuracy, but to verify that the proposed method transfers from simulation to hardware without fundamental modification.

4.5 Summary of Results

This chapter presented our results and evaluations of the proposed method in both simulations and limited hardware experiments. The simulations demonstrate the effectiveness of the hierarchical planning strategy and the coupled MPC, particularly in scenarios involving complex trajectories and interactions, directional changes, and higher accelerations. The hardware experiments further indicate that the proposed method can be executed on a physical system under real-world constraints.

While quantitative metrics provide insight into performance trends in simulation, many aspects of the system behaviour are best appreciated qualitatively. To facilitate a more intuitive understanding of both the simulated scenarios and the hardware experiments, recordings of representative tests are made available online at https://arian-kourangi.github.io/DA236X-Master-Thesis/.

Chapter 5

Discussion

In this chapter, we discuss the validity of the results obtained from the simulations and experiments, as well as key limitations and modelling choices underlying the proposed method. In Section 5.1 we interpret the results and assess the extent to which the intended behaviour is achieved. Section 5.2 then positions the proposed method relative to existing approaches and highlights the key trade-offs. The remainder of the chapter examines assumptions, limitations, and implementation aspects. Section 5.3 addresses system-level modelling assumptions and their implications for applicability in a microgravity environment, Section 5.4 discusses limitations of the hardware experiments, and Sections 5.5–5.7 focus on the offline planner, the online planner, and the MPC formulation, respectively.

5.1 Towards Untethered Transportation

The results presented in Chapter 4 suggest that the proposed method achieves the intended behaviour under the evaluated scenarios. Across all evaluated scenarios, the simulations indicate that collaborative, untethered transport of a passive object in a two-dimensional microgravity analogue environment is feasible using the proposed approach. The low pre-interaction velocity differences suggest that interactions between the robots and the object were gentle, while the low mean and variance of the final position deviation across scenarios indicate adequate precision in the final object placement across repeated trials. Furthermore, the experimental demonstration indicates that the proposed method remains feasible under non-ideal conditions, including environmental disturbances and unmodelled effects, even without any parameter optimization. This suggests a degree of robustness that supports the practical applicability of the approach and indicates its potential for enabling untethered, collaborative object transportation in realistic space-like environments. Further development and refinement of the method could improve cost-efficiency and scalability of transportation in orbital environments, potentially supporting debris-removal and related on-orbit operations.

5.2 Comparison with Existing Approaches

Comparing the proposed method to existing approaches highlights a clear trade-off between control authority and flexibility. Traditional space-based transportation methods, including docking mechanisms, tethered systems, and robotic manipulators, typically achieve high positional and velocity accuracy and allow for precise and predictable interaction with target objects. These systems also provide strong guarantees in terms of collision avoidance and operational safety through rigid or well-defined physical coupling.

In these respects, the proposed method is not competitive. By design, it sacrifices direct control authority over the object in favour of a contact-based, untethered interaction strategy. As a result, the achievable accuracy and predictability of the object trajectory are reduced, and the system becomes more sensitive to disturbances and modelling errors.

However, this loss of control authority enables increased flexibility and generality. Unlike grasp-based or tethered systems, the proposed approach does not require any dedicated mechanical interface on the transported object. This makes it applicable to a wider class of unactuated and uncooperative objects, such as space debris or legacy satellites not designed for servicing. Moreover, the absence of rigid connections simplifies system design and enables transportation without relying on purpose-built spacecraft.

Compared to impact-based approaches such as [14], the proposed method exhibits a different set of trade-offs. Impact-based strategies enable rapid and robust momentum transfer, which can lead to higher accuracy in achieving desired post-interaction velocities. In contrast, the prolonged contact interactions used in the proposed method introduce greater sensitivity to modelling inaccuracies, including friction, inertia, and positional misalignment, which can accumulate over the interaction duration and affect the final object trajectory. The primary advantage of the proposed approach lies in its reduced reliance on impulsive forces, lowering the risk of damage to both the robots and the transported object.

Overall, the proposed method is not intended to replace existing high-precision transportation techniques but rather to complement them. Its main advantage lies in enabling scalable and general-purpose transport of uncooperative objects without requiring mechanical coupling. As such, it is best suited for tasks where high precision and strict safety guarantees are not critical, such as debris removal or transportation in less constrained environments.

5.3 Validity, Assumptions, and Limitations

While the presented results indicate encouraging performance, several modelling assumptions and design choices limit the extent to which conclusions can be drawn regarding the behaviour of the proposed method in a fully three-dimensional microgravity environment. These assumptions introduce uncertainty when extrapolating the observed performance beyond the simulated setting. In the remainder of this section, we discuss the most significant of these assumptions and limitations and examine how they affect the validity and generalizability of the proposed method.

5.3.1 Model Similarity

The most fundamental limitation of the presented results concerns the similarity between the simulated environment and a true microgravity setting. The ATMOS test-bed and simulation environments constitute a two-dimensional microgravity analogue and, therefore, by design, can only approximate the physical conditions encountered in space.

A significant modelling simplification arises from the assumption of a frictionless environment. In the simulations, the robots, the transported object, and the supporting surface are all modelled as frictionless. This choice is made to avoid explicitly modelling a microgravity environment, which would substantially increase the complexity of the simulation setup. Consequently, frictional effects during contact between the robots and the object are neglected. In addition, exhaust gases produced by the robots' thrusters are not modelled, thereby neglecting their potential influence on the object during close-quarter maneuvers.

Overall, these modelling choices introduce some uncertainty regarding the applicability of the proposed method in a true three-dimensional microgravity environment. While the results support the feasibility of the approach as a proof-of-concept under idealized conditions, further investigation using more physically representative models and experimental validation would be required to assess its viability for real-world space operations.

5.3.2 Geometric and Kinematic Simplifications

A central modelling choice in the proposed method is the assumption that both the robots and the transported object can be represented as circular geometries. This deviates from realistic scenarios, in which satellites and debris typically exhibit irregular or complex shapes. This assumption is motivated by the need to simplify collision handling and contact modelling, as explicitly accounting for object orientation and shape-dependent interactions would introduce strong nonlinearities that are not tractable within the current MICP and QP frameworks. In addition, the method neglects the rotational dynamics of the transported object and considers only translational motion. These simplifications introduce ambiguity regarding how orientation control and angular dynamics would be handled in a realistic setting.

5.3.3 Measurement Assumptions and Noise

Another key assumption underlying the presented results concerns the availability and quality of state measurements for both the robots and the transported object. State information is obtained either directly from Gazebo or from a motion capture system, in contrast to space robotic systems, which rely on onboard state estimation. As a result, measurement noise, latency, and partial observability are negligible or entirely absent in the presented experiments. This constitutes a strong assumption, as sensing in realistic space applications is subject to noise, delays, and occlusions.

5.4 Hardware Experiments

Several characteristics of the experimental setup influenced the observed behaviour. In particular, imperfections and unevenness in the supporting surface of the ATMOS platform prevented the transported object from remaining stationary or following the straight-line trajectory that would be expected on a perfectly planar surface, which is required to approximate microgravity conditions. As a result, the observed object motion deviated from the nominal behaviour assumed in the simulation and planning models.

To mitigate this issue, we deployed a ”velocity-keeping” MPC on the object, activated only during non-interacting segments of its trajectory. This controller is designed to maintain the object's velocity at the value measured immediately prior to activation, effectively compensating for disturbances introduced by floor unevenness and friction. In particular, after an interaction when the robot releases the object, the object's direction and magnitude of motion are often significantly perturbed by imperfections in the supporting surface. The velocity-keeping MPC acts to preserve the intended post-interaction motion, approximating the behaviour that would be expected in a more faithful microgravity environment.

Another consequence of the limited test-bed size is the reduced temporal and spatial margin available for error correction. Since the transportation task must be executed within a confined workspace, both the online re-planner and the MPC have limited opportunity to compensate for deviations once execution has begun. Moreover, the constrained workspace necessitates operating at low velocities, increasing the relative influence of surface imperfections and disturbances on the object's motion.

Taken together, these factors limit experimental repeatability and highlight the need for a larger and more representative workspace to enable a more rigorous evaluation of the proposed method. Nevertheless, the experiments demonstrate that the planning and control framework can be executed on real hardware under constrained and imperfect conditions.

5.5 Offline Planner

A key limitation of the offline planning approach lies in the handling of task specifications using STL constraints. Notably, the final destination of the object is not explicitly included in the STL formulation. This omission was made because incorporating STL constraints directly into the MICP problem proved to be computationally prohibitive. Consequently, the final positions of objects are instead encoded directly in the MICP formulation, and STL constraints are applied conservatively. This limitation reflects a broader characteristic of MICP-based planning: as problem complexity increases, solver efficiency decreases rapidly, particularly for longer horizons and richer specifications.

Another significant challenge in the offline planner is controlling acceleration and force along planned trajectories. As discussed in Section 3.3.3.5, imposing constraints on accelerations or contact forces would introduce non-convex constraints that cannot be handled within the MICP framework. As a result, there exists a substantial abstraction gap between the offline planner and the actual system dynamics. In practice, this means that trajectories produced by the offline planner can include abrupt or nearly instantaneous velocity changes that are physically infeasible, leading to discrepancies between planned and executed behaviour.

5.6 Online Planner

One of the primary limitations of the proposed method lies in achieving the planned post-interaction object velocities. Because accelerations can only be influenced indirectly through attenuation, neither the offline planner nor the online re-planner can guarantee that the synthesized trajectories are dynamically feasible or that stable contact is maintained during interactions. This limitation is most pronounced in Scenarios 1–3, where higher interaction forces are required. In these cases, increased attenuation reduces the achievable post-interaction velocities, resulting in noticeable deviations from the velocities specified by the offline plan. By contrast, in Scenario 4, which involves substantially lower interaction forces and accelerations, the discrepancy between planned and achieved velocities is significantly smaller. A related consequence of these limitations is the relatively large deviation in task duration. The increased execution time arises from the same underlying challenges in accurately modelling and constraining accelerations across planning layers. As a result, the proposed method, in its current form, is ill-suited for time-critical operations that require strict temporal guarantees.

5.7 MPC

While Scenarios 1–3 demonstrate the potential benefits of incorporating a coupled MPC layer, caution is warranted when interpreting its performance and assessing its advantage over the nominal formulation. The primary objective of the coupled controller is to construct a continuously differentiable approximation of the coupled system dynamics. As a result, several modelling assumptions and simplifications are introduced, which limit the extent to which the observed improvements can be generalized beyond the simulation setting. In particular, the chosen activation function constitutes an approximation that inherently limits accuracy, as illustrated in Fig. 3.7, where its maximum value is bounded at 0.62. An alternative formulation based on hard activation thresholds using min–max operators was investigated but proved incompatible with gradient-based optimization.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

This thesis investigated the feasibility of cooperative, untethered transport of a passive object in a microgravity analogue environment. The primary objective was to evaluate whether a hierarchical planning and control architecture can reliably coordinate interactions, regulate momentum transfer, and achieve accurate object placement under idealized conditions. To this end, a combination of offline planning, online re-planning, and MPC was developed and evaluated in simulation and on a two-dimensional microgravity analogue test-bed.

Across all evaluated scenarios, the results demonstrate that the proposed approach is capable of achieving consistent and repeatable transport of a passive object, indicating the feasibility of our method as a framework for untethered, cooperative transport in microgravity.

A key insight from the experimental results is the dominant role of interaction quality in determining overall task performance. Deviations in object velocity following interactions were shown to propagate through subsequent phases of the task, directly affecting final placement accuracy. In this context, the integration of model predictive control plays an important role in mitigating interaction-induced errors. In particular, the coupled MPC formulation demonstrated clear advantages in scenarios characterized by higher interaction forces and more dynamically demanding object trajectories, where explicitly accounting for object dynamics improved velocity matching and reduced accumulated errors. Conversely, in scenarios involving lower accelerations and gentler interactions, the nominal MPC achieved comparable performance, indicating that the benefits of coupled control are regime-dependent.

Despite these encouraging results, the conclusions of this work are subject to several important limitations, as discussed in Chapter 5. Consequently, the presented results should be interpreted as demonstrating feasibility under controlled conditions rather than robustness in operational environments.

Overall, this work establishes that cooperative, untethered transport of passive objects in microgravity analogue environments is feasible within a hierarchical planning and control framework. While the presented results do not establish robustness or operational readiness, they provide clear evidence that interaction-based cooperative transport is a viable approach and a promising foundation for further research into coordinated transport in microgravity.

6.2 Future Work

Several directions for future work naturally follow from this study. Most notably, extending the method to three-dimensional motion, incorporating rotational dynamics and more realistic contact and friction models, and explicitly accounting for sensing noise and estimation uncertainty would significantly improve physical fidelity. From a planning perspective, alternative formulations capable of handling non-linear dynamics, force constraints, and actuator limits may help reduce the abstraction gap between planning and execution. Additionally, relaxing the assumption of circular robots and objects would allow the method to generalize to agents and payloads with more complex geometries. Finally, more extensive experimental validation under less restrictive assumptions would be essential to assess the robustness, scalability, and practical applicability of the proposed approach.

6.3 Reflections

6.3.1 Economic Aspects

The application of untethered transport in microgravity presents potential economic benefits for the space industry, particularly by increasing the efficiency of transporting un-actuated objects. By enabling flexible and reconfigurable transport operations, such systems may reduce reliance on purpose-built spacecraft and dedicated control architectures. This may lower development and operational costs and reduce the financial barriers to entry for in-space logistics and servicing missions.

6.3.2 Environmental Aspects

Improving the efficiency of transportation in space is closely linked to the long-term sustainability of space activities. Reusable robotic platforms capable of manipulating and relocating objects in orbit may decrease the need for single-purpose spacecraft. This could reduce the number of launches required to support logistical operations, leading to lower propellant consumption and a reduced environmental impact associated with launch activities.

In addition, the presented method is well suited to address the growing challenge of space debris. Autonomous robots capable of untethered transport and relocation of uncooperative objects could support debris mitigation and on-orbit servicing efforts. By facilitating the removal or controlled de-orbiting of space debris, this technology could contribute to a safer and more sustainable orbital environment for future missions.

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Appendix A

MICP Constraints

In addition to the constraints presented in Section 3.3, we ensure collision avoidance between robots and between objects (but not between robots and objects) with

$$\begin{aligned}
&\|p_{S_i}(t) - p_{S_j}(t)\|_2 \leq \text{rad}_{S_i} + \text{rad}_{S_j}, \quad \forall t \\
\iff &\begin{cases} \text{box}(r_{S_i}^{(k)}) \oplus B_\infty^{\text{rad}_{S_i}} \ \cap \ \text{box}(r_{S_j}^{(l)}) \oplus B_\infty^{\text{rad}_{S_j}} = \emptyset, \\ \forall k, l \in \mathbb{N}_S \end{cases} \\
\iff &\ h^{(k)} \cap h^{(l)} \neq \emptyset.
\end{aligned} \tag{A.1}$$

where B_∞^{rad} denoted the infinity-norm ball with radius rad. Furthermore, we impose a mutual-exclusion constraint such that each robot Bézier segment can be involved in at most one interaction overall

$$\sum_{o \in \mathcal{O}} \sum_{j=0}^{N_O-2} z_{(R,O)}[k,j] \leq 1 \qquad \forall R \in \mathcal{R}, \ \forall k \in \mathbb{N}_{[0,N_R-1]} \tag{A.2}$$

Similarly, we enforce that after any object Bézier segment involved in an interaction, the next object segment must be interaction-free.

$$\sum_{R \in \mathcal{R}} \sum_{k=0}^{N_R-2} \Big(z_{(R,O)}[k,j]) + z_{(R,O)}[k,j+1] \Big) \leq 1, \quad \forall O \in \mathcal{O} \ \forall j \in \mathbb{N}_{[0,N_O-2]} \tag{A.3}$$

Lastly, we enforce that interactions along an object trajectory cannot occur consecutively with the same robot. That is, if an object interacts with a robot at two distinct segments, then at least one interaction with a different robot must occur in between. To formulate this constraint, we introduce the binary variable $Y_{[R,O,j]}$ which indicates whether object O interacts with robot R on object Bézier segment j, irrespective of the specific robot segment involved. This variable is defined as

$$Y_{[R,O,j]} = \sum_{k=0}^{N_R-2} z_{[R,O]}[k,j] \quad \forall R \in \mathcal{R},\ \forall O \in \mathcal{O},\ \forall j \in \mathbb{N}_{[0,N_O-2]} \tag{A.4}$$

Using this auxiliary variable, we impose the following constraint to prevent consecutive interactions with the same robot

$$Y_{[R,O,j_1]} + Y_{[R,O,j_2]} \leq 1 + \sum_{\substack{\tilde{R}\in\mathcal{R} \\ \tilde{R}\neq R}} \sum_{j=j_1+1}^{j_2-1} Y_{[\tilde{R},O,j]} \quad \begin{array}{l} \forall O \in \mathcal{O},\ \forall R \in \mathcal{R} \\ \forall j_1, j_2 \in \mathbb{N}_{[0,N_O-2]} \text{ s.t } j_1 < j_2 \end{array} \tag{A.5}$$

Appendix B

MPC Cost Function

B.1 Nominal

To construct the cost function for the nominal controller, we first interpolate the Bézier curves at each shooting stage and use them to generate the reference trajectory given in (3.24). At every shooting stage, the tracking error is defined with respect to this reference, and the corresponding stage cost is written as

$$\begin{aligned} e_s &= [p_R, v_R] - [p_R^{ref}, v_R^{ref}] \\ e_q &= 1 - \left(q_R^\top q_R^{ref}\right)^2 \\ J^*(\tilde{x}(k)) &= e_s^\top Q_r e_s + e_q^2 Q_q + u^\top R u \end{aligned} \tag{B.1}$$

B.2 Coupled

For the coupled controller, the reference trajectory is augmented to include the object, using the object reference defined in (3.26). The tracking error is defined over both the robot and the object states:

$$\begin{aligned} e_s &= [p_R, v_R, p_O, v_O] - [p_R^{ref}, v_R^{ref}, p_O^{ref}, v_O^{ref},] \\ e_q &= 1 - \left(q_R^\top q_R^{ref}\right)^2 \end{aligned} \tag{B.2}$$

In addition to the state and orientation tracking terms, we penalize tangential contact forces in order to discourage sliding during interaction. The tangential projection operator and the corresponding force error term are defined as

$$
\begin{aligned}
t &= I_3 - \vec{n}\vec{n}^\top \\
\vec{F}_t &= t\vec{F}_R \\
e_t &= \eta \, \|\vec{F}_t\|_2^2
\end{aligned}
\tag{B.3}
$$

The complete stage cost for the coupled controller is therefore

$$
J^*(\tilde{x}(k)) = e_s^\top \, Q_r \, e_s + e_q^2 \, Q_q + u^\top R u + Q_t \, e_t \tag{B.4}
$$